Weaponizing Interdependence: A Dynamic Game of U.S. Semiconductor Controls and Chinese Critical-Mineral Restrictions

Executed computational methods and a source-traceable empirical feasibility study

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Abstract

Economic coercion may reduce present access while changing the target's incentives and ability to develop substitutes. We implement separate rare-earth, semiconductor and policy-observability research components and a bounded four-state reciprocal game. The executed package archives primary records and extracts 87 versioned rare-earth numerical observations, 22 additional annual production observations, a documented subset of 26 projects, 27 semiconductor legal-event phases and 8 corporate investment-input observations. These records demonstrate industrial development, heterogeneous project states and the importance of separating product forms, legal access and production capability. They do not identify a population completion model, a complete China-free material network or a causal adaptation response. A conserving multi-commodity linear program, competing-risk machinery, opacity coverage rules, differential equations and constrained game solvers pass 477 executed tests. Hypothetical finite-grid game candidates satisfy reported residual and unilateral-deviation checks in selected parameter regions; negative-response fine-grid failures and nonzero off-grid residuals prevent continuous-state equilibrium certification. A one-sided analytical benchmark verifies self-eroding leverage conditional on positive induced adaptation. A reciprocal constant-commitment extension satisfies sufficient restricted uniqueness bounds in nine hypothetical scenarios; it does not establish feedback or unrestricted open-loop equilibrium. Nine original Entity List schedules also yield 488 printed entry versions, preserving mixed review standards and distinct legal clocks without inferring access loss. Real-world threshold probabilities, a headline historical opacity index and national welfare conclusions remain unavailable. The contribution of this draft is an inspectable implementation, carefully bounded descriptive evidence and an explicit account of what further observations would make the substantive hypotheses testable.

Keywords: economic coercion; rare earths; semiconductor controls; material-flow optimization; dynamic games; historical information sets.

Release status. This is an evidence-limited research draft, not a submission-ready empirical paper. Synthetic validation and hypothetical strategic scenarios are kept separate from empirical results in every data directory and figure caption. All numerical claims below are linked to generated outputs and original-source manifests. The supplied blueprint defines methods; its assertions are not treated as independently verified data.

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1 Introduction

The economic value of a chokepoint depends on both current dependence and the evolution of alternatives. Semiconductor controls and rare-earth restrictions provide an asymmetric setting in which to investigate this distinction. An equipment or software restriction can alter access immediately, while a new separation facility or a qualified process technology takes time to develop. Restrictions may accelerate substitution through investment and policy support, impede it by removing essential inputs, or have little effect. The sign of induced adaptation is consequently an empirical question.

This research asks whether using a chokepoint today can reduce its future coercive value. It separates three measurements before integration: physically deliverable rare-earth magnet services without Chinese inputs or processing; Chinese domestic semiconductor capability versus usable foreign access; and observable policy-information conditions. A reciprocal dynamic game connects independently constructed state variables without treating industrial tonnage, legal severity or research spending as national utility.

The present execution resolves substantial computational requirements and establishes selected descriptive facts. It also reaches concrete evidentiary limits. Public company reports provide useful operating histories and native-product output, but do not supply the complete element-level recipes, grade demand, inventories and routing rights necessary to estimate China-free service. Primary legal records establish rules and dates, not delivered technology access or indigenous mastery. The missing observations are retained as missing, rather than replaced by plausible-looking country scores.

The scope of this draft is therefore narrower than the proposed final empirical study. It reports an executed methodology, an auditable source corpus, a project discovery ledger that includes unsuccessful states, and numerical scenario results. It does not infer that Chinese leverage ends in a particular year, that sanctions cause successful technological substitution, or that observed nondisclosure is deliberate. The incomplete results are useful restrictions on the claims that the available evidence permits.

2 Literature review

Farrell and Newman (2019) explain how asymmetric network positions can create surveillance and chokepoint power under appropriate jurisdictional and institutional conditions. This study takes that mechanism as a starting point rather than claiming to originate weaponized interdependence. Clayton et al. (2024) formalize the integration–security trade-off and the possibility of excessive fragmentation. Their original working-paper text was retrieved; the version identifies a December 2024 paper revised in April 2025.

Rare-earth differential games already exist in Zou et al. (2022). Woods (2026) develop an opacity-signaling mechanism and illustrative diversification simulations. The intended distinction here is the combination of traceable project histories, physical process constraints, observable policy coding and independently assessed numerical equilibrium diagnostics. This execution does not establish that the complete empirical contribution has yet been achieved.

Semiconductor-related firm evidence is not uniformly evidence of successful substitution. The publisher-deposited abstracts of Anwar et al. (2024), Hu et al. (2024) and Hayakawa and Yang (2026) describe different innovation and R&D outcomes and estimation designs. They concern inputs

or innovation indicators rather than the complete qualified production target specified here. We do not convert their estimates into manufacturing-capability elasticities. Full-text access remained incomplete for several papers; the citation audit distinguishes verified bibliographic registration, abstract reading and original-text reading.

Investment under uncertainty motivates investigating project delays (Bloom, 2009; Gulen and Ion, 2015; Handley and Limão, 2015; Dixit and Pindyck, 1994). Duration and competing-risk methods distinguish failure, completion and censoring (Cox, 1972; Fine and Gray, 1999); heterogeneous treatment timing motivates cohort-specific causal designs when their assumptions can be defended (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021). Proper scores evaluate predictive distributions rather than the appearance of smooth forecasts (Gneiting and Raftery, 2007; BRIER, 1950). Stationary game computation draws on the distinction between Markov-perfect strategies, finite stochastic-game existence and numerical approximation (Maskin and Tirole, 2001; Fink, 1964; Pakes and McGuire, 2001; Bardi and Capuzzo-Dolcetta, 1997). None of these references establishes empirical identification or pure-feedback equilibrium existence for the particular model implemented here.

3 Institutional and historical background

3.1 Two different industrial dependencies

The IEA's report summary, originally published on 8 April 2026, reports approximate Chinese shares of 2024 magnet-rare-earth mining, refining and sintered permanent-magnet production of 60%, 91% and 94%, respectively. The retrieved live summary and dated landing page are archived separately. Later navigation on the live site is excluded, and the page is not asserted to be a certified historical web snapshot. These aggregate shares motivate a stage-specific investigation; they do not determine an importing bloc's feasible material flows.

The semiconductor dependency is a technology and access network. The registered layers are EDA/design IP, lithography, deposition/etch/implantation, metrology and process control, foundry integration, HBM where required, and advanced packaging. A license requirement can allow specified transactions; jurisdiction, end use, technical classification and exceptions matter. A product-level chronology is more informative than a timeless embargo indicator.

3.2 Original-rule chronology

The selected primary corpus contains U.S. Federal Register texts and metadata, BIS releases and Dutch legal instruments. The October 2022 U.S. rule has separate equipment, U.S.-person and computing implementation phases (Bureau of Industry and Security, 2022a). The January 2026 licensing revision illustrates why publication and first public availability must be separated: the reviewed document was filed for public inspection on 13 January, and published and effective on 15 January (Bureau of Industry and Security, 2026a). These are event phases, not 27 independent policy shocks.

Dutch national authorization requirements are coded separately from U.S. rules. Japanese and Chinese native legal-source coverage remains incomplete. The blueprint's assertions about April 2025 Chinese measures and later suspensions remain acquisition candidates where the authoritative text could not be retrieved. No completed historical legal finding is inferred solely from a numbered

announcement in the blueprint or from a company's interpretation of law. Nor is the selected U.S./Dutch event corpus presented as a fully consolidated legal regime through the freeze.

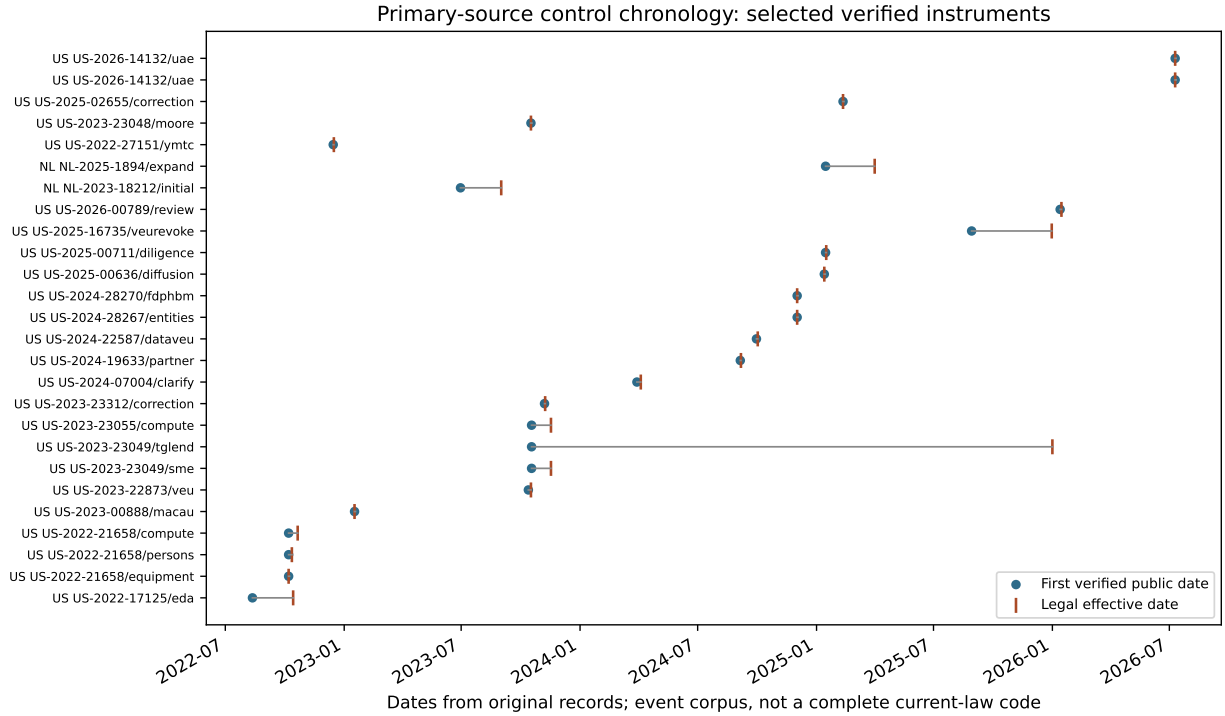


Figure 1: Selected verified primary legal-event phases. Circles and bars distinguish verified public and effective dates. The corpus is incomplete and is not an estimated physical access-loss series.

4 Theoretical framework

4.1 Physical adequacy and dynamic adaptation

A feasible service threshold is a statement about delivering final products under specified exclusions and demand. It is not a bifurcation. Let $Y_t \in [0,1]$ be the weighted served share from the material network. A separate reduced-form proxy satisfies

$$\dot{r} = \eta I(1-r) - \delta r, \qquad \eta > 0,\ I \geq 0,\ \delta \geq 0. \tag{1}$$

Writing $A = \eta I$ and $\lambda = A + \delta > 0$, its unique steady state is $r^* = A/\lambda$ and

$$r(t) = r^* + (r_0 - r^*)e^{-\lambda t}. \tag{2}$$

For $r_0 < h < r^*$, the upward crossing time is

$$T_h = \lambda^{-1} \log \frac{r^* - r_0}{r^* - h}. \tag{3}$$

Equality $r^* = h$ gives asymptotic approach, not finite crossing. If $A = \delta = 0$, every initial state is a neutral equilibrium; the quotient formula must not be evaluated. Starting below one, finite

constant effort does not yield $r(t) = 1$ at finite time. A finite network can nevertheless serve 100% of a quarter's demand, because r and Y are distinct objects.

A genuinely nonlinear auxiliary model is

$$\dot{z} = \kappa\{\Lambda(\theta + \beta z) - z\}, \quad \Lambda(v) = (1 + e^{-v})^{-1}, \quad \kappa > 0. \tag{4}$$

Its folds require $\beta > 4$ and occur at $z_\pm = (1 \pm \sqrt{1 - 4/\beta})/2$, with $\theta_\pm = \log[z_\pm/(1 - z_\pm)] - \beta z_\pm$. The outer roots are stable when $\beta z(1 - z) < 1$. At $\beta = 4, \theta = -2$, the single root $z = 1/2$ is nonhyperbolic and locally asymptotically stable through a cubic term. These are conditional mathematical properties, not evidence of industrial hysteresis.

4.2 Falsifiable mechanisms

The registered hypotheses concern project implementation, restriction-induced investment, immediate semiconductor access, delayed functional capability, opacity-related delay or overreaction, endogenous network feedback, self-eroding leverage and reciprocal strategic interaction. Negative and zero induced adaptation are allowed. An increase in announcements without qualified output does not confirm physical resilience; an increase in R&D without a functioning target stack does not confirm capability substitution.

5 Data and identification

5.1 Source architecture and two clocks

Every admitted numerical observation has a source ID, immutable artifact digest, product basis, native unit, event or accounting-period date, verified information-public date and extraction locator. Raw revisions are preserved. As-of selection first restricts public availability and then selects the latest eligible version. It never makes a later revision visible at an earlier origin. A target announcement can be known before the target date without constituting a completed event.

SEC filing dates provide conservative verified availability for observations extracted from filings. They are not proof that an earlier earnings release did not exist. Interval dates remain intervals, and document signatures without a verified release date are not silently promoted to publication times. The Saltend on-hold evidence is retained with an unavailable publication-date field and excluded from historical as-of estimation. A mistaken attempted company identifier and pre-merger SPAC records are explicitly rejected in the adjudication log.

The rare-earth discovery ledger is purposive and issuer-weighted. It covers 26 projects and 36 activities in the curated subset, with further candidates awaiting original documents. It contains suspensions, development, construction and commissioning cases; it is not an exhaustive statistical population. Large numbers of repeated KPI disclosures or quarterly rows do not create independent project-transition events.

5.2 Executable information and original-source checks

An additional information-set audit uses the actual 37-claim finance/material table. Selecting only on event dates would admit 15 unavailable claims at 31 December 2025 and 12 at 30 June 2026. Applying the source-public clock removes them; these counts diagnose look-ahead and do not

measure forecasting accuracy. Four annual-report release clocks remain unresolved after rejecting 18 authoring-metadata, page-date and filename candidates. In the archived Pensana reports page, the FY2023–2025 title links identify distinct PDFs, while their download arrows all point to the FY2023 PDF. No report identity or public date is inferred from the misleading arrow link.

The extraction path now hashes one immutable original-byte snapshot and processes that same snapshot, including PDF extraction through standard input. Editable text aids cannot supply claim evidence. A separate recheck verifies 2783 quotations and quarterly table cells across 24 datasets and 60 originals, enforcing explicit PDF page locators. Passing these checks establishes support in the present archived originals. It does not certify independently archived release-time bytes, a globally earliest public date, or independently measured industrial performance.

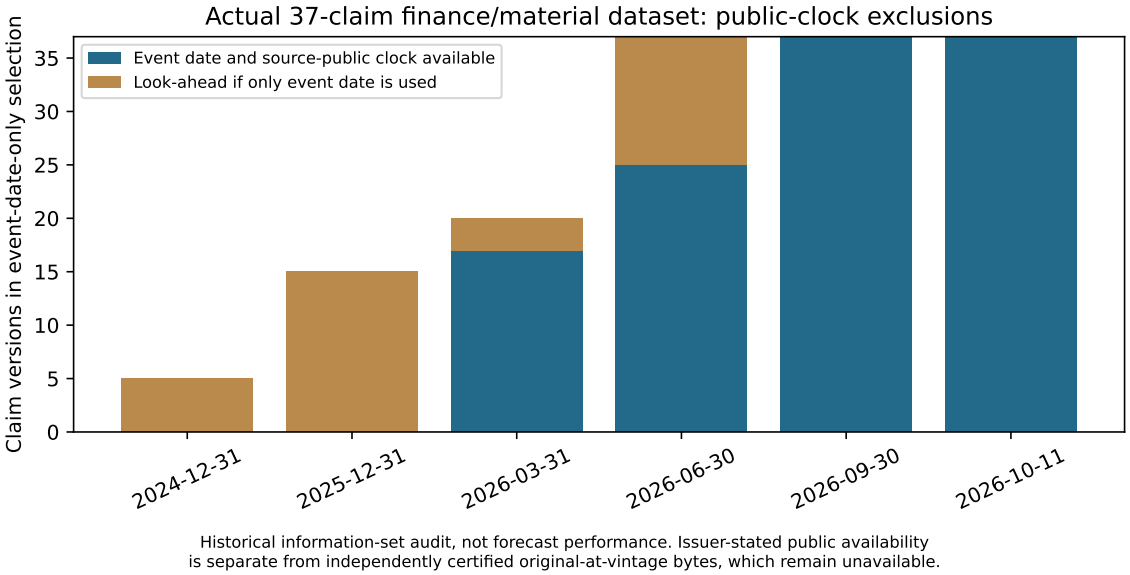


Figure 2: Actual claim-version exclusions under the two-clock rule. The upper bar segment is unavailable at that origin despite an earlier event date. These are information-set diagnostics, not forecast scores or evidence of deliberate opacity.

5.3 Identification boundaries

Eleven supplied PDFs now cover all eleven previously unavailable document requests. Seven are browser-print copies of SEC documents created on 22 October 2026; four are supplied native-format PDFs. Identity, printed URLs where available, page counts, checksums and historical document clocks are verified separately. The four quarterly SEC filing clocks are re-derived from archived submission bytes. Incorporated technical reports use conservative public-by dates from the dated parent filing links, rather than their effective or signature dates. SMIC first-page datelines and the legal documents' publication/filing notices supply the other clocks. This admits source-reported statements while explicitly withholding original-at-vintage byte certification. It does not convert the failed HTTP attempts into successful downloads. The Donald report uses the full incorporated exhibit URL in the parent filing, not its nonunique exhibit filename. The available evidence does not identify population completion hazards, a full physical service series or national adaptation elasticities. The semiconductor corporate panel has one firm's reported inputs and lacks a credible

untreated comparison set, frozen product-level exposure and independently verified target-specific capability. No causal event-study coefficients are estimated. The absence of qualifying public evidence is treated as unverified, not zero domestic capability.

Opacity regressions require a coverage-valid lagged index, project risk sets, finance/support histories and separate technical and geopolitical uncertainty controls. Those conditions are not satisfied. Strategic payoff scales are not identified from observed legal actions. All numerical game payoff and adaptation primitives are hypothetical scenarios, explicitly registered in the parameter table.

6 Rare-earth supply-chain results

6.1 What the primary records establish

MP Materials reports 200, 1,294 and 2,599 metric tonnes of finished and packaged NdPr oxide in calendar 2023, 2024 and 2025 (MP Materials Corp., 2025, 2026a). Its total REO-in-concentrate measure has a different product basis, and its definition includes downstream commissioning feed from 2023. Adding these series would double count material. The 2025 report describes late-year magnet-equipment commissioning; projected expanded magnet capacity is not the same as demonstrated qualified shipments.

The Lynas issuer archive supplies annual reports for 2016–2026 and selected quarterly reports. Reported NdPr product-family output is 5,655 tonnes in fiscal 2024, 6,558 in fiscal 2025 and 7,260 in fiscal 2026 (Lynas Rare Earths Ltd., 2026). Fiscal years end in June. These observations are neither identical accounting windows nor guaranteed identical product specifications to MP's calendar-year oxide output. No cross-company sum or conversion to contained elemental mass is used as an empirical service estimate.

An additional extraction retains 66 source-vintaged quarterly KPI cells, representing 30 distinct quarter–product observations. All six complete fiscal-year/product sums for 2024–2026 exactly reconcile with their own annual reports. This checks internal accounting consistency, rather than independent production measurement. The 23 Lynas milestone claims span four projects and seven activities. Including the additional downstream disclosures below, the documented roster contains 70 records, 26 projects and 36 activities. It remains a selected discovery roster, not a population risk set.

The White Mesa disclosure reports 38 tonnes of separated NdPr in 2024, alongside nominal capacity of 800–1,000 tonnes per year (Energy Fuels Inc., 2025). Its subsequent report describes common crack/leach resources shared with uranium processing, and combined Phase 1/Phase 2 expansion totals (Energy Fuels Inc., 2026a). An empirical LP must encode these shared constraints and must not add a combined expansion total to its included first phase.

The project-state review identifies substantive non-success cases. Bear Lodge permitting was suspended in 2016 and later restarted; Upton is a demonstration operation using stockpiled sample, with up to 10 native tons expected over a finite campaign rather than a perpetual annual commercial rate (Rare Element Resources Ltd., 2026a). Vara Mada/former Toliara retains a reported 2019–2024 suspension history under a stable project identity (Energy Fuels Inc., 2026a). Stillwater's first-quarter 2026 commissioning involved running and testing equipment, with orders expected subsequently; the aggregate Less Common Metals rate includes other critical minerals and cannot all be assigned to NdPr (USA Rare Earth Inc., 2026).

Iluka's 1,200-tonne magnet-oxide offtake quantity covers an initial four-year term beginning in 2028,

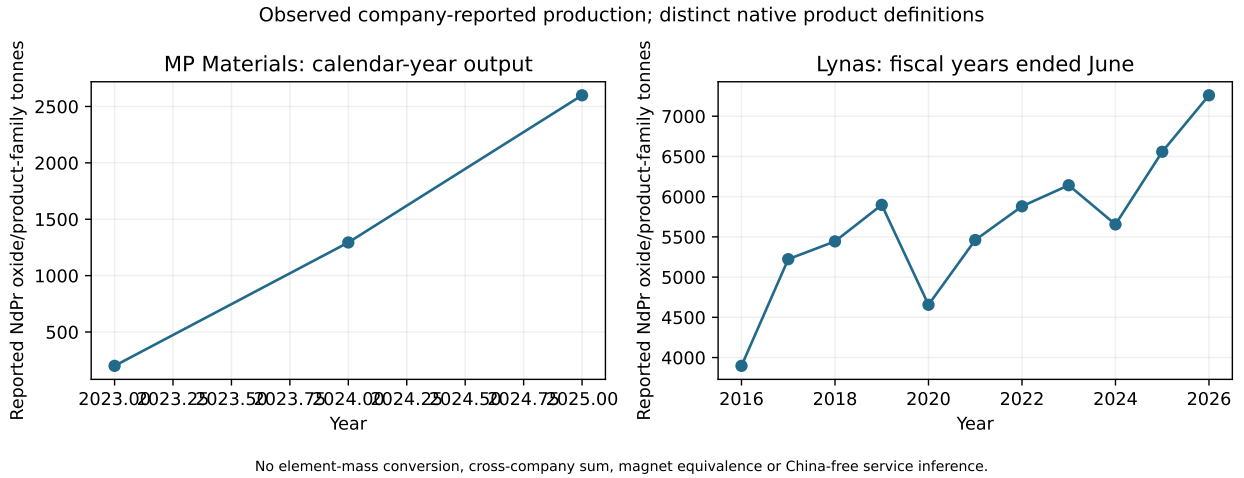


Figure 3: Actual issuer-reported native-product output. Panels retain distinct fiscal calendars and product definitions. Output growth does not establish complete China-independent processing, magnet qualification or demand coverage.

not one year's capacity. Its projected product mix depends on additional feedstocks (Iluka Resources Ltd., 2026a). Pensana's dated construction notice reports ongoing Longonjo construction and a 2027 commissioning target (Pensana PLC, 2026a). These findings demonstrate why event type, native basis and conditional feed must be read directly from sources.

Lynas' FY2026 report explicitly states that its planned Texas heavy-rare-earth facility will no longer proceed and describes funding reallocated to oxide purchases (Lynas Rare Earths Ltd., 2026). Earlier reports' planning, funding, target and wastewater-permitting disclosures remain separate vintages; the exact cancellation decision date is unverified. Existing archives also distinguish Kalgoorlie's December 2023 first feed, June-quarter 2024 first MREC shipment and subsequent operational ramp (Lynas Rare Earths Ltd., 2024a,b). First separated dysprosium and terbium production was reported for May and June 2025, respectively (Lynas Rare Earths Ltd., 2025b); these milestones do not supply independent qualified-throughput or recovery measurements. Two documented Australian–Malaysian material links retain unknown route capacities. Finally, MP's 2022 filing retrospectively reports Mountain Pass cold-idling in mid-2015 and a July 2017 production relaunch (MP Materials Corp., 2023). Those observations become usable only at their verified 2023 filing clock; they cannot be inserted into earlier-origin forecasts.

6.2 Downstream disclosures and a finite qualified-batch claim

Additional filings identify a planned 3,750-tonne-per-year metal/alloy facility in Lacq, France and a distinct planned metal activity at Stillwater (USA Rare Earth Inc., 2026). The same filing mentions Caremag's commissioning target, which is counterparty evidence rather than independent operating confirmation. Energy Fuels describes ASM's Korean Metals Plant as producing rare-earth metals and alloys while the proposed acquisition remains conditional (Energy Fuels Inc., 2026a); neither its completed ownership transfer nor its stable qualified rate is inferred. Lynas discloses a non-binding Malaysian magnet-project MoU in FY2025 and a later July 2026 partnership with JS Link, with prospective 3,000-tonne annual capacity. Its Vietnam framework and convertible-instrument agreements with LS Eco Energy do not establish plant FID or commissioning (Lynas Rare Earths

Ltd., 2025a, 2026). Persistent asset identities prevent counting successive announcements or adjacent stages twice.

The Energy Fuels filing also reports that approximately 1.2 metric tonnes of White Mesa NdPr oxide were processed into 3.0 tonnes of permanent magnets by an unnamed South Korean counterparty, with automotive QA/QC qualification claimed (Energy Fuels Inc., 2026a). It identifies Florida and Georgia monazite as the oxide feed origin and describes an announcement dated 9 September 2025; the archived filing's verified availability is 26 February 2026, so earlier-origin forecasts cannot use this extraction. The source-reported product-mass ratio is approximately 2.5, not an elemental recovery. Aggregate mass accounting implies

$$M_{\text{other inputs}} = M_{\text{magnets}} + M_{\text{other outputs}} - M_{\text{oxide}} \;\geq\; M_{\text{magnets}} - M_{\text{oxide}} \approx 1.8 \text{ tonnes}, \tag{5}$$

conditional on the approximate reported quantities and nonnegative other outputs. This is a necessary accounting bound, not an identified chemical recipe. Additional rare-earth and non-REE inputs, their origins, processing recoveries, production dates and independent customer documents remain missing. The historical consumed batch is not opening inventory, annual capacity, sustained qualified throughput or complete China-free service. The Korean counterparty is not equated with ASM's plant merely because both are in Korea. A separate 45H block-grade statement and 99.9% pilot Dy-purity claim remain distinct from the quantified batch and from process yield.

6.3 MP: production, sales and reporting conventions

Three original annual filings supply 174 native table-cell versions: 36 annual volume KPIs, 96 quarterly volume KPIs and 42 monetary inventory balances, accompanied by 19 source-linked definitions (MP Materials Corp., 2024, 2025, 2026a). Comparative vintages remain separate. The extraction retains 38 N/A cells and two displayed dashes; neither becomes a measured physical zero. Of 24 quarter–annual comparisons, 14 are numerical identities, six are consistently N/A and four lack fully numerical support. The latter involve early N/A quarters before reported separated-product production; no quarterly value is imputed to manufacture an identity. Twelve monetary component identities also reconcile.

In calendar 2025 the issuer reports 50,692 metric tonnes of REO-content production and 8,922 of concentrate sales; NdPr production is 2,599 tonnes of finished and packaged oxide, while sales are 1,994 oxide-equivalent tonnes. These are distinct industrial and accounting objects. Upstream REO production includes material fed to downstream circuits from the second quarter of 2023, overlapping separated production. NdPr sales can represent oxide or metal and include Materials-to-Magnetics intercompany sales from the fourth quarter of 2025. Neither adding the stages nor treating all sales as external qualified deliveries is valid.

The FY2023 report illustrates sales measurement with a hypothetical 1.25 oxide-equivalent/metal mass ratio, whereas FY2024 and FY2025 state an assumed 1.20 ratio. The newly supplied Q2 2026 filing explicitly dates an actual change from 1.20 to 1.25 to Q1 2026 and states that prior periods have not been recast (MP Materials Corp., 2026b). This resolves the latest adoption quarter without turning the earlier hypothetical illustration into an observed convention or measured chemical recovery. Transit, conversion timing and actual versus assumed yields can separate production from reported sales. The 2025 NdPr KPI difference of 605 tonnes therefore does not identify an inventory increase. Reported year-end total inventories of USD 252.099 million, including USD 39.622 million of current finished goods, are mixed book values, not physical oxide or magnet stocks. The filing supplies no Magnetics operating KPI; that absence is not evidence of zero production.

The Q2 filing reports 840 tonnes of finished, packaged NdPr oxide for April–June 2026 and 1,757 tonnes for January–June, versus 597 and 1,160 in the respective prior-year periods. REO-in-concentrate output is 11,072 and 24,055 tonnes, a different and overlapping stage. NdPr sales of 1,006 and 2,012 oxide-equivalent tonnes retain the updated convention and include intersegment transfers. The printed cash-flow line corroborates the separate SEC-facts observations of USD 307.711 million and USD 59.473 million first-half PPE cash outflows in 2026 and 2025. This is a line-level financial reconciliation, not an independently re-extracted XBRL-context or physical-throughput identity.

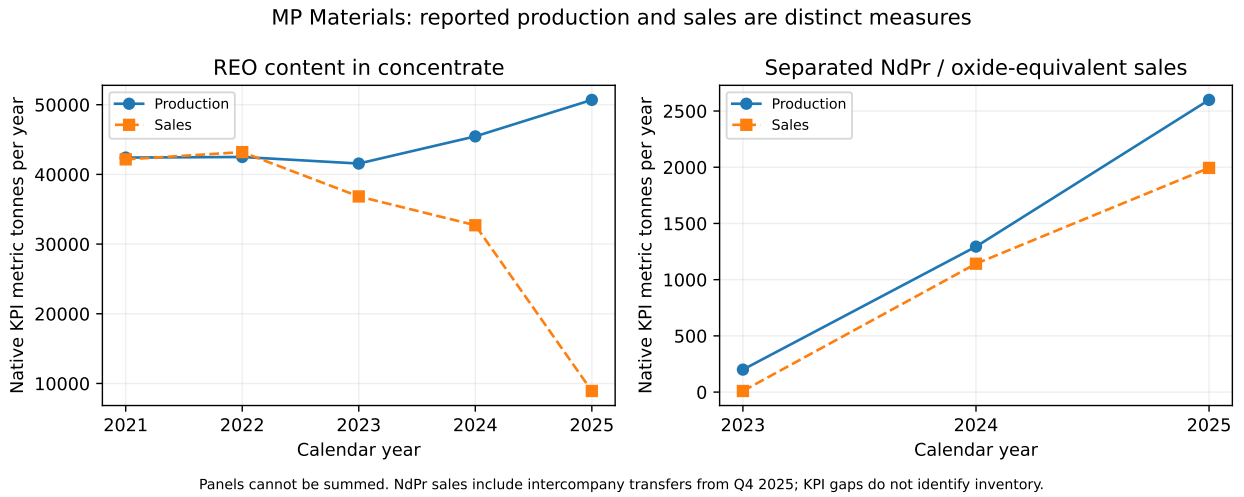


Figure 4: Actual issuer-native production and sales KPIs. Upstream and downstream measures overlap; NdPr sales use an oxide-equivalent reporting convention and include internal transfers from Q4 2025. Differences do not establish physical inventories, recovery or final service.

6.4 Commercial routes, input links and ownership versions

The source-vintaged commercial extension adds 33 original claims from the MP FY2023–2025 filings (MP Materials Corp., 2024, 2025, 2026a). The two earlier reports describe VREX Holdco's Vietnam rare-earth oxide-to-metal processing activities and ownership of 49% by MP and 51% by Shenghe Singapore, an indirect majority-owned subsidiary of Shenghe Resources Holding. They assign day-to-day management to Shenghe and state that MP does not control the board. The reported equity shares close to 100% in both snapshots. These are ownership/management disclosures, not a complete legal-control determination or verification of every input's nationality. A Vietnam location alone cannot establish a fully China-independent route.

The FY2025 report states that MP sold its entire 49% interest in May 2025. It does not identify the resulting complete ownership, plant closure, or termination of all conversion relationships. The same report states that sales to China ceased in July 2025 and separately describes Shenghe reagent/material procurement. These facts distinguish a sales destination from input origin: cessation of China sales cannot certify that all upstream inputs were China-free. The reported procurement totals cover the period when Shenghe was considered a related party; they do not identify purchase timing relative to the July sales cessation. Nominal procurement disclosures are financial values, not physical chemical quantities or input-origin assays. The five comparative-year groups preserve repeated source vintages rather than manufacturing independent observations. Q4 2025 internal

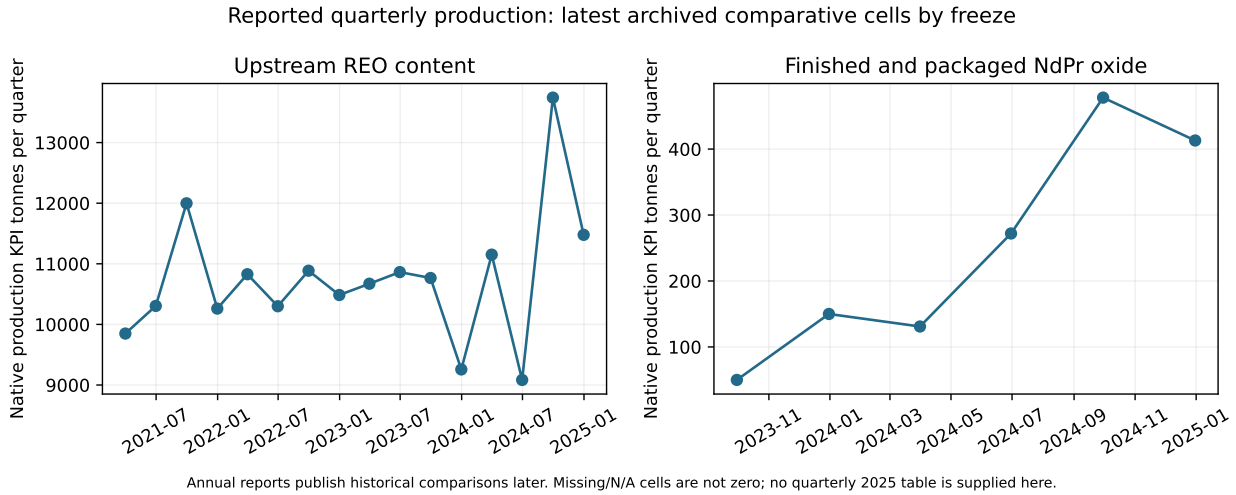


Figure 5: Quarterly MP production cells from retrospective annual-report comparisons. Every cell retains its report-publication clock. N/A cells remain missing; this extraction contains no quarterly 2025 KPI table.

oxide supply from Mountain Pass to Independence is a separately reported link, not a completed end-to-end magnet-service recipe.

Event	Reported interval	Selected original public date
MP equity acquisition	Dec 2023	2024-02-28
MP equity disposal	May 2025	2026-02-26
China sales cessation	Jul 2025	2026-02-26
Internal oxide feed	Oct–Dec 2025	2026-02-26

Table 1: Issuer-reported event intervals and selected original-source public clocks. The intervals are not exact event days, certified earliest disclosure dates or estimated reporting-delay observations.

The five admission cases separate processing location from a registered Shenghe supplier-group exclusion sensitivity. They retain missing input origins, finite volume, qualification, ownership at the forecast origin, recipes/recoveries, route rights/throughput and grade demand. None admits a complete empirical route. The screen does not solve an empirical LP, determine national legal control, impute zero real supply or estimate service probabilities.

6.5 White Mesa: native flows, stocks and conditional capacity

An additional within-issuer panel extracts 49 claims from the FY2023–2025 filings, conservatively available on 23 February 2024, 26 February 2025 and 26 February 2026 (Energy Fuels Inc., 2024, 2025, 2026a). The FY2023 report describes 184 tonnes of monazite processed and 95 tonnes of contained TREO in RE carbonate in 2022, followed by 326 and 260 tonnes, respectively, in 2023. The product/feed ratios are approximately 0.5163 and 0.7975. These compare bulk feed with contained TREO; unknown assays, opening/closing process inventories and other flows prevent interpreting them as chemical recoveries. The earlier 2021 statement uses 400 native tons of monazite and 120 tonnes of TREO. Its rounded glossary relation of 1.102 tons per tonne does not establish an exact ton standard, so no precise metric ratio is inserted.

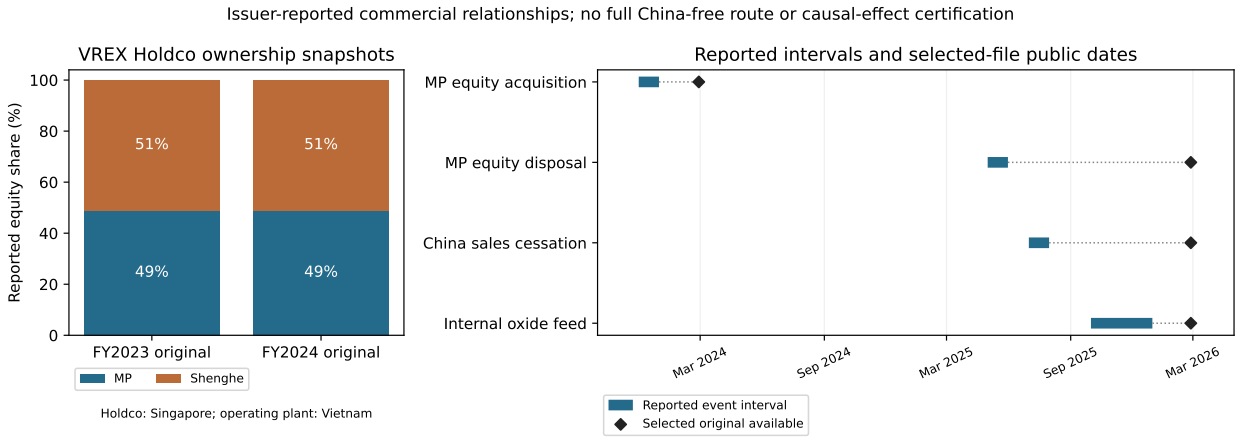


Figure 6: Actual reported ownership snapshots and documentary clocks. The May 2025 disposal becomes usable from the selected 2026 original's public clock; no ownership trajectory is interpolated between snapshots.

The two later filings both describe a 2024 commissioning run processing 500 tonnes of monazite and producing 38 tonnes of NdPr product. The FY2024 Properties passage calls that product *oxalate*; the FY2025 passage calls it *oxide*. Both descriptions survive with their own public clocks. They may reflect different process stages or an oxide-equivalent reporting convention, but the available passages do not adjudicate the mass basis. The equal numerical ratio 38/500 does not establish equal chemical product masses or a recovery of 7.6%. Three earlier research labels for the generic output/nameplate observations are therefore corrected from assumed oxide to unconfirmed reported product form, preserving every native amount and source date. This is a documented research coding correction, not a fabricated issuer revision.

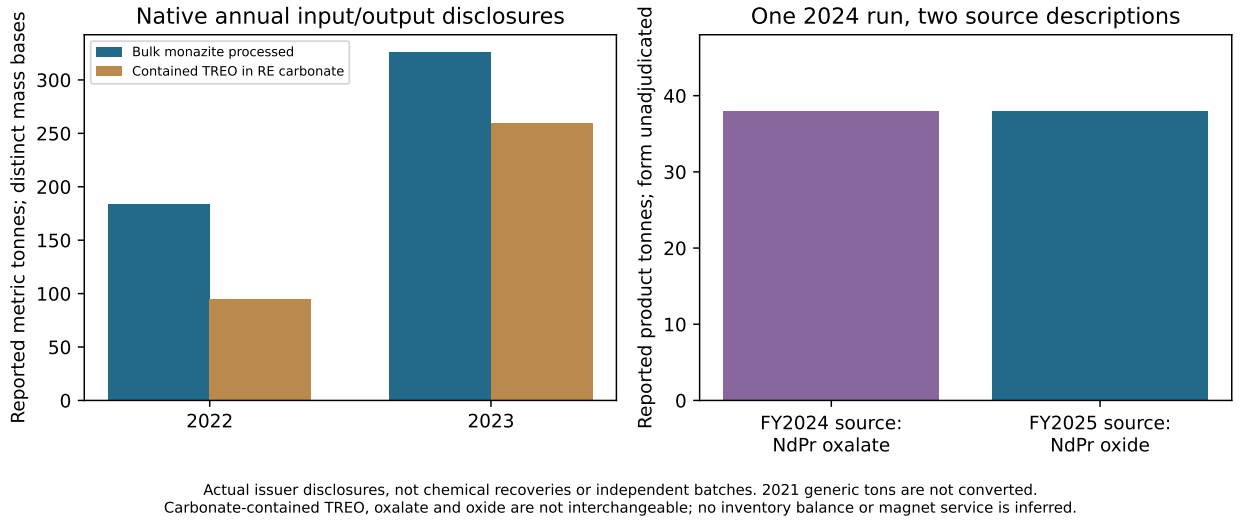


Figure 7: Actual source-reported feed/output quantities with different material bases retained. The right panel repeats one described 2024 episode across source versions; it does not add two independent batches or convert oxalate mass to oxide mass.

The FY2025 filing reports approximately 34 tonnes of separated NdPr inventory explicitly at 31 December 2025, and 1.7 tonnes of sampling sales during 2025. Its additional 26 tonnes of NdPr and four tonnes of Sm+ RE carbonate in solution are work in process; the sentence does not independently date those measurements. The FY2024 report separately describes 28 tonnes of NdPr and four tonnes of Sm+ in solution. These are neither finished stocks nor estimates of subsequent output. Historical production cannot substitute for opening inventory, sampling sales cannot close an otherwise unobserved stock-flow ledger, and stock changes cannot be inferred across unspecified measurement dates. Across 20 actual stock/origin combinations, no verified current opening stock is admitted without the missing date/form and intervening-flow evidence. This does not establish zero physical inventory.

The shared crack/leach front end permits only one feed, uranium or monazite, at a time. This instantaneous restriction allows annual time sharing only when processing rates and schedules are supplied; licensed uranium-output pounds are not a monazite-feed rate. The expected expansion table separately reconciles NdPr $1{,}049 + 5{,}513 = 6{,}562$, Tb $12 + 48 = 60$ and Dy $35 + 165 = 200$ native table tonnes per annum. The Phase 1 HREE components are planned, and an existing-NdPr label does not independently verify its expected rate. Combined totals must not be added again, and product totals must not be pooled across elements or chemical forms.

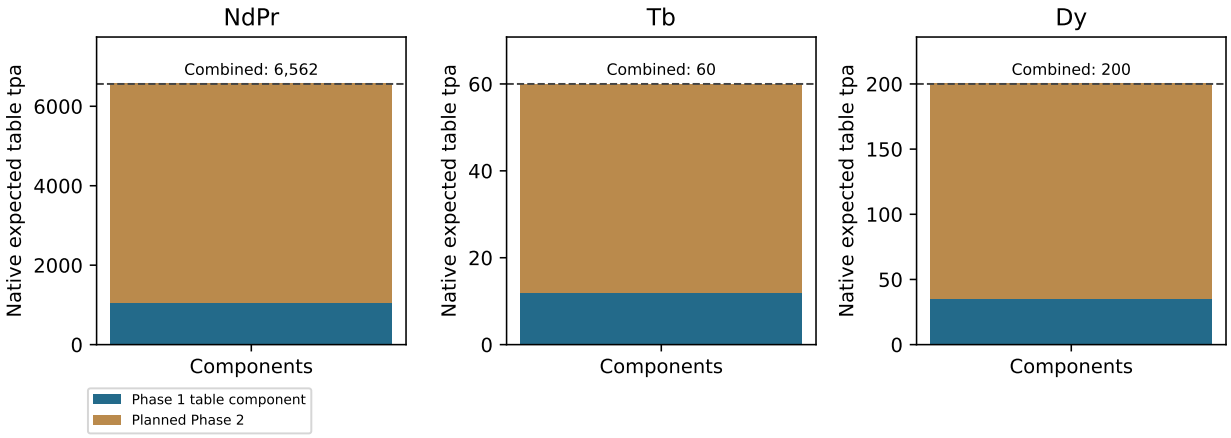


Figure 8: Issuer expected capacity components and their combined totals. These conditional plans are distinct from observed throughput, validated recoveries, stock availability and delivered magnet service.

Finally, the FY2025 engineering discussion describes full commissioning in Q2 2024 and completion of the initial commercial run in Q3; its accounting discussion places construction and commissioning completion in Q3. Those event definitions still require reconciliation. The supplied Vara Mada technical report adds oxide-composition evidence discussed below; the Donald technical report now supplies the separately conditioned constraints reported below.

6.6 Donald: nested material bases and a documented missed target

The Donald technical report is effective 31 December 2025 and conservatively public by 26 February 2026 through the exact incorporated link in Energy Fuels' annual filing (Datamine Australia Pty Ltd (Snowden Optiro), 2026; Energy Fuels Inc., 2026a). The supplied 217-page browser copy was created after the freeze; neither the effective date nor the copy date is substituted for public availability. Its Table 1.3 reports a total reserve HM grade of 4.5% and explicitly states that individual oxide grades are percentages of that HM component. Oxides are contained within the mineral assemblage, not additional material. Table 15.2 separately provides pit-optimization recovery assumptions to REEC. For a declared finite ore quantity F and common throughput q, these imply the necessary conditional constraints

$$0 \le q \le F, \qquad d_e \le q\, h\, a_{e|\mathrm{HM}}\, r_e\, \phi_e, \quad e \in \{\mathrm{Nd}, \mathrm{Pr}, \mathrm{Dy}, \mathrm{Tb}\},$$

where h is the bulk-ore HM mass fraction, $a_{e|\mathrm{HM}}$ the oxide share of HM and ϕ_e the contained-element fraction of its oxide. Treating an HM-nested oxide grade as a bulk-ore grade would inflate the coefficient by $1/0.045 \simeq 22.2$. These reserve-average and pit-design assumptions do not estimate commercial recovery or certify a matched operating batch.

Element	Oxide (% HM)	Recovery (%)	Element (kg/t ore)
Nd	0.2	92	0.07099
Pr	0.056	92	0.01919
Dy	0.041	88	0.01415
Tb	0.0064	88	0.002155

Table 2: Donald conditional reserve/pit-design coefficients. Common HM grade is 4.5%; source recovery assumptions stop at mine concentrate. Source precision limits interpretation of derived digits. No available ore, downstream separated output or magnet service is estimated.

Sixteen diagnostic finite-feed linear programs, including zero feed, throughput limits, closed routes and owner removal, match analytical bounds and conserve all four elements. Diagnostic ore normalizations are stored separately from empirical outputs. The 293 Mt reserve is not admitted as an available stock; the proposed 30-tonne product-bin capacity is not inventory. The source's 1,000 dry tonnes/hour times 7,500 assumed operating hours reconciles to 7.5 Mt/year design throughput, not an achieved flow.

The financial model's Table 22.5 instead reports REEC oxide shares of total TREO: 13.85% Nd, 3.85% Pr, 2.67% Dy and 0.41% Tb, with 60.6% TREO in concentrate. These yield conditional composition on the as-stated product basis; moisture comparability is not independently certified. The pit-costing section's 8% moisture assumption is not silently applied to that grade. Vara Mada's comparative Donald distribution is different (15.90, 4.44, 2.04 and 0.36%, respectively). Both have the same conservative public-by date but different disclosed contexts; they are not matched samples or evidence of a historical grade change. Similarly, a pilot result of 91% rare-earth *mineral* recovery and 56% TREO grade cannot be combined with the later financial composition as one observed operating recipe. The confirmatory product's 92–94% rare-earth-phosphate statement describes purity, not recovery.

The contracts and costing distinguish REEC shipment toward Seattle/White Mesa from HMC coproduct processing in China. REEC offtake starts after FID and may be suspended below a price floor, allowing third-party sales. At the technical-report date, a new REEC export licence was

being sought. The later filing asserts all major regulatory approvals; that broader statement does not independently audit every secondary permit, route capacity or input origin. Thus neither a China-bound HMC coproduct nor the proposed U.S. REEC destination establishes the independence or dependence of every input in the eventual rare-earth chain.

A genuine dated expectation can also be evaluated. Table 22.8 planned financing/FID for March 2026. The Q2 filing explicitly states that no FID had occurred by 30 June and gives a revised Q3 2026 target (Energy Fuels Inc., 2026b). Consequently the original target was missed by at least 91 elapsed days through 30 June, but this negative-event observation enters the research information set only on 5 August. The revised target is not an observed FID, and the 91-day lower bound is neither a completed delay duration nor a population hazard estimate. Reported Energy Fuels equity is 9.48% at the technical-report effective date and 12.7% at June 2026; the 49% full earn-in remains conditional.

6.7 Supplied technical and quarterly evidence

The Mountain Pass technical report's Table 10-9 reports pilot-informed overall recovery estimates of 89.7% for grouped PrNd, 97.9% for SEG+, 74.9% for lanthanum and 8.9% for cerium (SRK Consulting (U.S.), Inc., 2026). The report explains the use of legacy plant data and bench/pilot activities to design and size circuits. These values corroborate planning assumptions rather than identify a quarterly operating recovery. Its mixed-product specification, approximately 75% Nd_2O_3 and 25% Pr_6O_{11} with a stated tolerance, is a product recipe rather than a feed assay.

Vara Mada Table 19-6 reports Toliara-monazite oxide-distribution percentages of 18.60 Nd_2O_3, 5.20 Pr_6O_{11}, 0.50 Dy_2O_3 and 0.20 Tb_4O_7 (Energy Fuels Inc., 2026c). Their denominator is the rare-earth oxide distribution, not raw monazite mass. Applying formula-specific atomic weights gives conditional contained-element fractions per tonne of total REO; it supplies neither feed availability nor an operating recovery. The report also states that the identified exploitation permit did not then include monazite. This is a dated technical-report statement; subsequent permitting is not inferred.

The latest Energy Fuels filing preserves shared Phase 1 crack/leach limits and adds a material design-version issue (Energy Fuels Inc., 2026b). Its page 54 prose describes planned additions of 20 tpa Tb oxide and 120 tpa Dy oxide, while its page 56 table lists Phase 1 heavies at 14 and 48 tpa. Within that table, Phase 1 plus Phase 2 sums exactly to 6,229 NdPr, 80 Tb and 288 Dy tpa. These differ from the earlier filing's planning totals. Both the within-document and historical versions are retained; no selected total is promoted to achieved output or added to the earlier plan. Product scope, design revision and timing require issuer or engineering adjudication.

USA Rare Earth's quarterly filing reports Wheat Ridge demonstration operations beginning 15 June 2026, followed by a 14 July announcement of recycled-swarf oxide samples and expected downstream qualification at Less Common Metals (USA Rare Earth, Inc., 2026). Recycling returns are not new external feed. The filing distinguishes selection for DOE award negotiations from an executed funded award and states that Commerce agreements had yielded no disbursement by 30 June. It reports the TMRC acquisition closed on 7 August, after the reporting endpoint but before the filing became public. These are different industrial and financing events; none establishes stable qualified magnet deliveries.

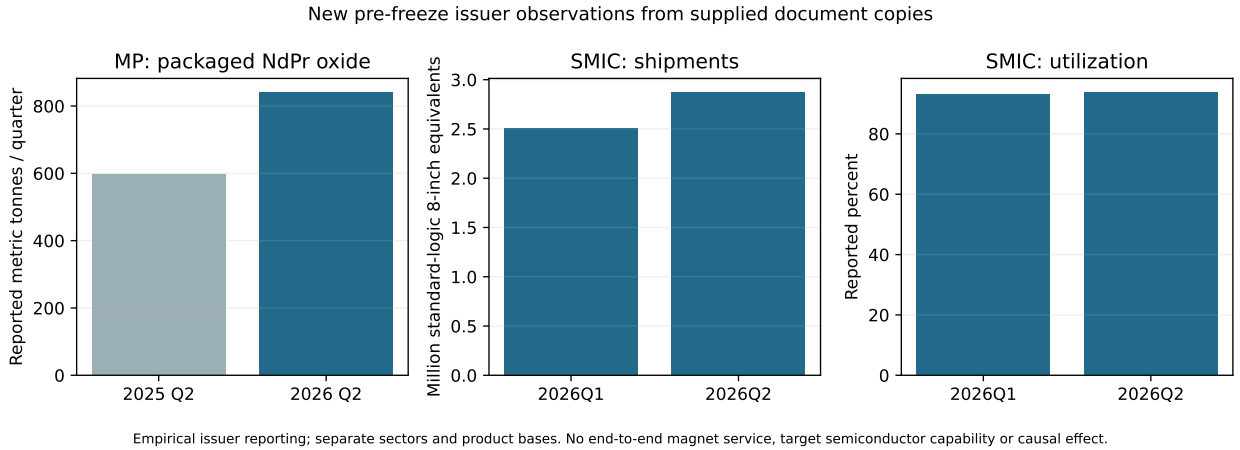


Figure 9: New issuer-reported observations from supplied copies. Panels retain distinct sectors, units and product bases. Neither a magnet-service series nor a target semiconductor-capability score is inferred.

6.8 Consolidated cash investment inputs

The archived SEC company-facts representation supplies 53 filing-vintaged cash-payment records across 28 distinct reported durations from 2021 onward. The measure is reported cash payments to acquire property, plant and equipment, in nominal USD; the exact SEC taxonomy tag is retained in the data dictionary. First-half 2026 payments are USD 307.711 million, compared with USD 59.473 million in first-half 2025. Where direct quarter tags exist, all three available 2024–2026 first-half reconciliations satisfy Q1 plus Q2 equals H1. These are internal consistency checks within the same SEC representation.

Reported durations and original filing clocks remain explicit. Comparatives retain their later publication vintages; the fiscal year of the reporting document does not replace the observation's start/end dates. Pre-2021 combination/predecessor periods are excluded, and overlapping quarterly, year-to-date and annual records are not summed together. The latest quarterly filing texts could not be downloaded through the attempted endpoints, so original XBRL-context reconciliation remains incomplete. These consolidated nominal cash inputs cannot be allocated to individual projects or manufacturing stages, interpreted as delivered output, or attributed causally to restrictions, subsidies or opacity.

6.9 Finance, publication clocks and feedstock-dependent product plans

An additional extraction records 37 financial, reserve, contractual and prospective-product claims. 4 report clocks are linked to exact issuer records: Pensana's construction and interim reports to articles published 14 January and 27 March 2026, and Iluka's half-year report and offtake notice to dated issuer cards of 19 August and 23 June 2026. The notice itself is dated 22 June. A date elsewhere on the page, a filename, a document signature or a page-modification timestamp is insufficient to date another linked report. Four archived annual reports consequently retain unresolved public clocks. The dated issuer associations support conservative availability; original-at-vintage web/PDF bytes and a globally earliest disclosure date are not independently certified.

Pensana's unaudited report reconciles July–December 2025 PPE additions of USD 10.773620 million

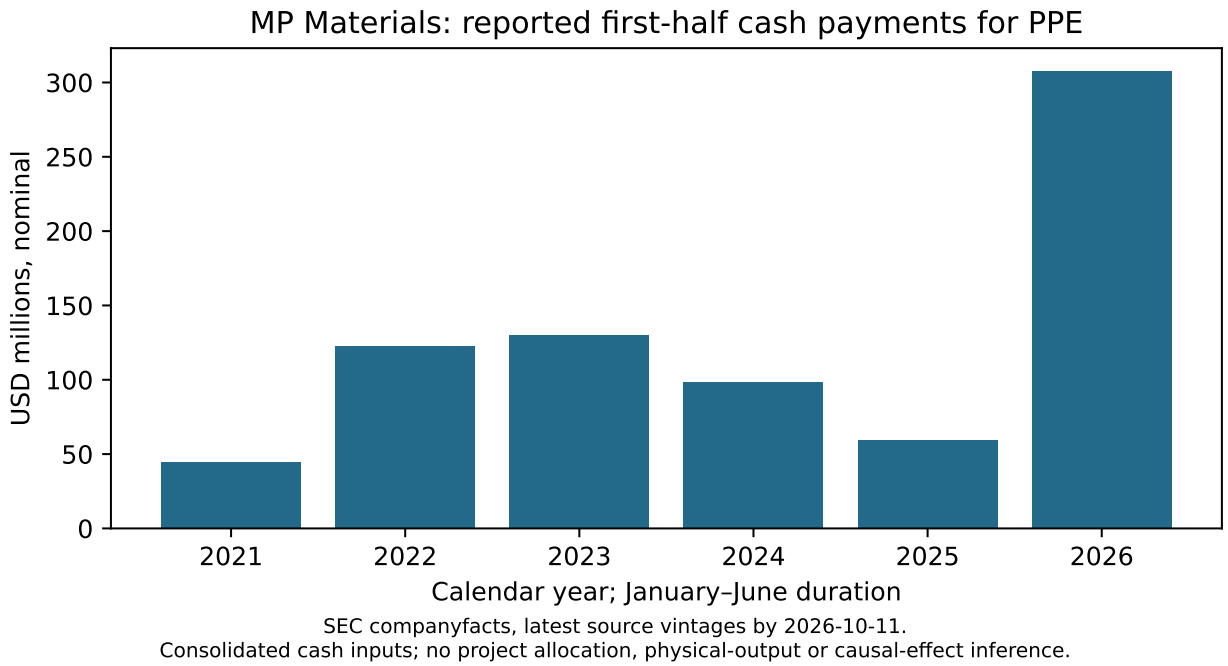


Figure 10: Actual tagged first-half cash payments for property, plant and equipment in the archived SEC company-facts data. Nominal consolidated inputs; no plant allocation, physical service or causal response is inferred.

and Saltend intangible additions of USD 0.225102 million to total additions of USD 10.998722 million. A capital-working-capital adjustment of USD 4.999892 million reconciles these additions to USD 15.998614 million in cash payments. The comparative 2024 amounts reconcile independently (Pensana PLC, 2026b). These are consolidated financial objects with explicit signs and periods, rather than a project-specific measure of completed physical capacity. The March 2026 proposed USD 165 million Cascade transaction remains subject to documentation and share applications in that report; its going-concern disclosure still depends on additional funding. Neither an announcement nor a geological reserve is treated as received project cash or available processing inventory. The report dates main Longonjo construction to July 2025, retaining March 2026 as this extraction's source-public clock.

Iluka reports AUD 897.5 million drawn and AUD 752.5 million undrawn against a base AUD 1,650 million EFA facility at 30 June 2026, with a separate undrawn AUD 75 million overrun allowance. The comparative base-facility balances reconcile as well. The report confirms the offtake access condition while retaining a draw sequence requiring initial funding arrangements to be used first. Eneabba's AUD 236.7 million half-year PPE additions include capitalised interest and accruals; they are not the change in loan balances or cash payments (Iluka Resources Ltd., 2026b). Contractual commitments remain obligations, and currencies and accounting objects are never aggregated across these issuers.

Iluka's prospective stockpile-only figures are approximately 3,500 tonnes of NdPr and 170 tonnes of combined Dy/Tb, versus 5,500 and 750 with additional feedstocks, subject to assemblage (Iluka Resources Ltd., 2026a). On a commensurate same-product basis, the ratios are 63.6% and 22.7%. The quoted footnote does not explicitly supply an annual denominator; none is inserted. Combined

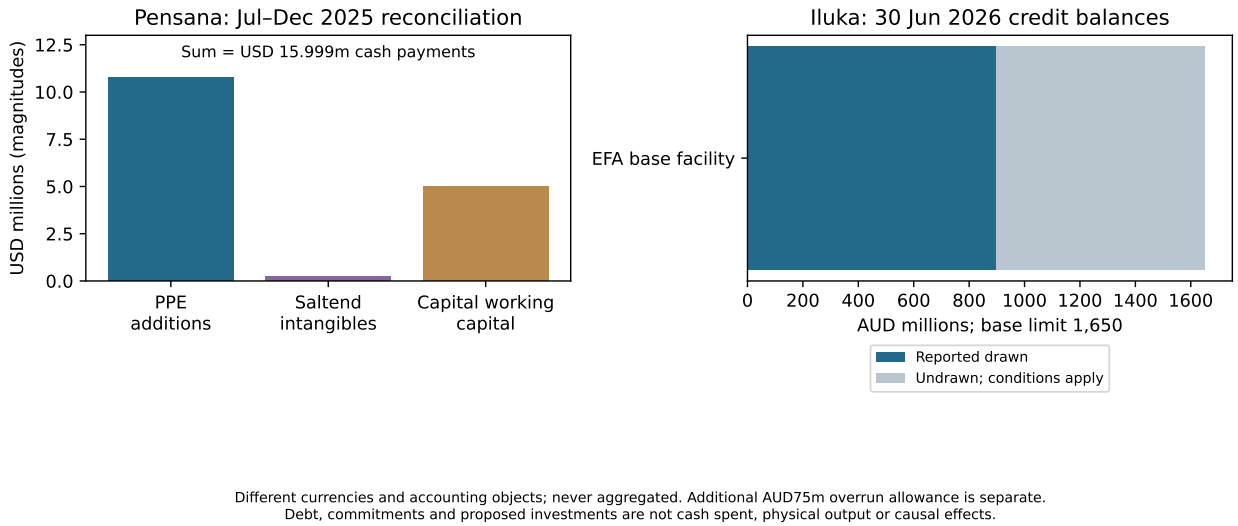


Figure 11: Issuer-reported accounting categories in different currencies. Pensana's additions plus capital-working-capital adjustment reconcile to cash payments; Iluka's base facility separates drawn and undrawn balances. Internal accounting consistency is not independent transaction verification, a financing-completion probability or physical output.

Dy/Tb is not split into separate elements, no recovery or demand is inferred, and these comparisons are not the LP service fraction y. The 1,200-tonne four-year offtake remains a future combined-product contract; the footnote places 90% of material under take-or-pay. Its minimum-price revenue calculation is conditional on product sales, not received or unconditional cash.

6.10 Source-calibrated concentrate constraints

The two archived MP annual filings supply 32 additional claims: 28 numerical engineering inputs and four categorical disclosures (MP Materials Corp., 2025, 2026a). The source's PrNd denotes the combined group labelled NdPr here; no elemental split is implied. The reported average REO composition is 15.7% PrNd, 1.8% SEG+, 32.3% lanthanum and 50.2% cerium, which sum to one joint feed distribution. These are shares conditional on REO content, not shares of bulk concentrate or independently available output streams. Resource- and reserve-pit optimization notes assume concentrate containing 60% TREO and a PrNd-oxide separation recovery of 89.7%. These are source-calibrated planning inputs; operating recoveries have not been estimated.

Conditional on these bases, a necessary grouped-oxide constraint is

$$0 \le d_{\mathrm{NdPr,REOeq}} \le 0.60\,(0.157)\,(0.897)\,q = 0.0844974q, \qquad 0 \le q \le F, \tag{6}$$

where q is native concentrate tonnes and F must be separately measured finite available feed. The result is an oxide-equivalent group bound, not a measured sale-product mass, an Nd/Pr elemental split, an annual rate or a complete production activity. Four source/context coefficient records and 12 normalized finite-feed LP checks retain unknown industrial stocks and downstream demand. The zero-feed case produces zero output.

The reserve-model Ce-recovery assumption changes from 11.5% to 8.9% between the FY2024 and FY2025 reports, while its La assumption changes from 75.0% to 74.9%. The corresponding resource-model assumptions remain unchanged. Both source versions and contexts are retained. These

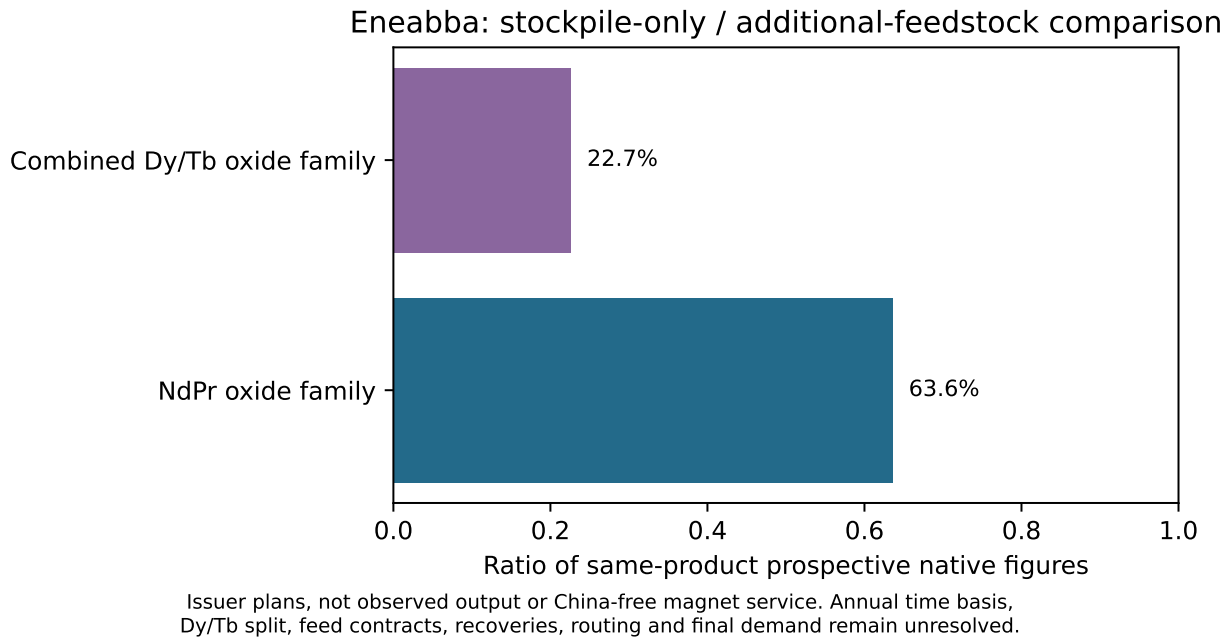


Figure 12: Ratios of prospective issuer product figures under two feedstock cases. These measure stated plan dependence on additional feed, not observed output, China-independent magnet demand coverage or a complete multistage outage calculation.

differences cannot establish deterioration in actual plant recovery or an adaptation response. The resource-estimate HREE omission is an estimate-scope statement, not evidence of zero HREE in feed. It does not identify Dy or Tb flows.

6.11 Milestone revisions and observation requirements

Three archived Rare Element Resources filings yield 15 source-checked history observations. The FY2024 report places substantial Upton demonstration-plant construction completion in late 2024, whereas FY2025 places it in late 2025 (Rare Element Resources Ltd., 2025, 2026a). Both describe commissioning beginning in November 2024; the later report adds a subsequent pause, review and rework before plant operations formally commenced in early 2026. The disjoint completion-year accounts require event-definition and rework adjudication. They do not establish that either disclosure is erroneous, and the later statement cannot overwrite the earlier information vintage.

The FY2023 filing scheduled plant operation for mid-2024; FY2024 expected mid-2025 (Rare Element Resources Ltd., 2024, 2025). These targets remain separate from realized demonstrations and independently qualified commercial operation. The FY2024 report explicitly pairs the total planned campaign of 10 tons with 9.1 tonnes of NdPr oxide. This supplies a rounded metric target, without establishing the exact ton standard, achieved production, annual capacity, recovery or a replenishable supply stream. Native "mid" and "late" year phrases use full-year outer enclosures; "early 2026" is enclosed by January 1 and the verified source date. These windows are neither uniform distributions nor confidence intervals.

The supplied Q2 2026 report narrows the formal demonstration start to March 2026 after an April 2025 design review and related rework (Rare Element Resources Ltd., 2026b). It separately

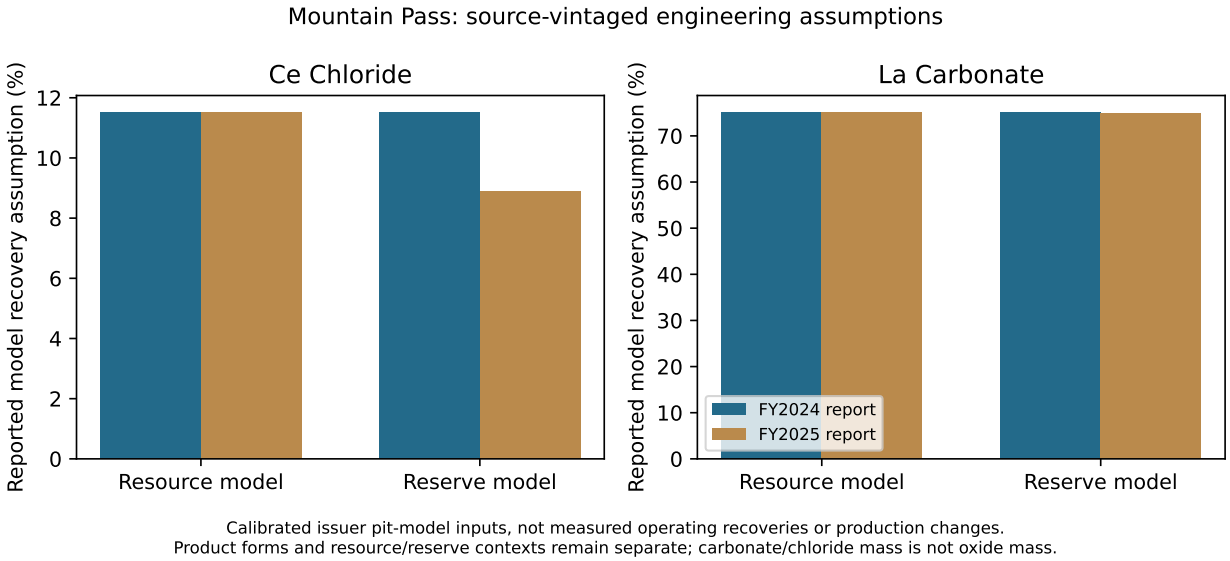


Figure 13: Source-calibrated engineering assumptions from the two original annual filings. Resource and reserve contexts, as well as carbonate/chloride product bases, remain separate. These are issuer pit-model assumptions, not observed operating recoveries.

reports subsequent equipment/feed-concentrate delays and expects full end-to-end processing in Q3 2026 for up to twelve months, producing up to ten native tons over the campaign. March is therefore not coded as verified full-process or commercial operation; the Q3 expectation remains a forecast at that source vintage. This refines event definitions without resolving the conflicting substantial-construction years or creating a first-exit surveillance spell.

Across the full discovery roster, 34 adjacent observation pairs include 20 same-state reobservations and 4 pairs with unobserved intermediate states or potential definition changes; 15 lack an event window at one or both endpoints. An allowed endpoint pair does not establish a first exit: intervening suspension, re-entry and reporting delay may be unobserved. The observation audit therefore admits no first-exit exposure spells. Its 44 quarterly source-disclosure origins end at October 11, including an explicitly partial final quarter. Physical projects-at-risk and known-survival exposure remain missing. The counts characterize this selected archive, not a population hazard or policy-opacity measure. Discovery selection uses the earliest source clock visible in the available archive, independently of row insertion order and retrospective event dates; it does not identify a project's true announcement or state-entry date.

6.12 Physical optimization and its present empirical limit

The implemented process network has activities q_a, commodity flows F_{hk}, closing inventories Z_{vk} and final service deliveries d_g. It maximizes $\sum_g w_g d_g / D_g$, subject to activity capacities, admissible transport and resource limits, finite external supply and material balances:

$$Z_{vkt} = Z_{vk,t-1} + M_{vkt} + \sum_{h \to v} F_{hkt} - \sum_{v \to h} F_{hkt} + \sum_{a \in v} (d_{ak} - b_{ak}) q_{at} - L_{vkt}. \tag{7}$$

Joint-product coefficients share an activity variable. Molecular-weight conversion and process losses are distinct. Registered disposal handles genuine coproduct surpluses explicitly; no free

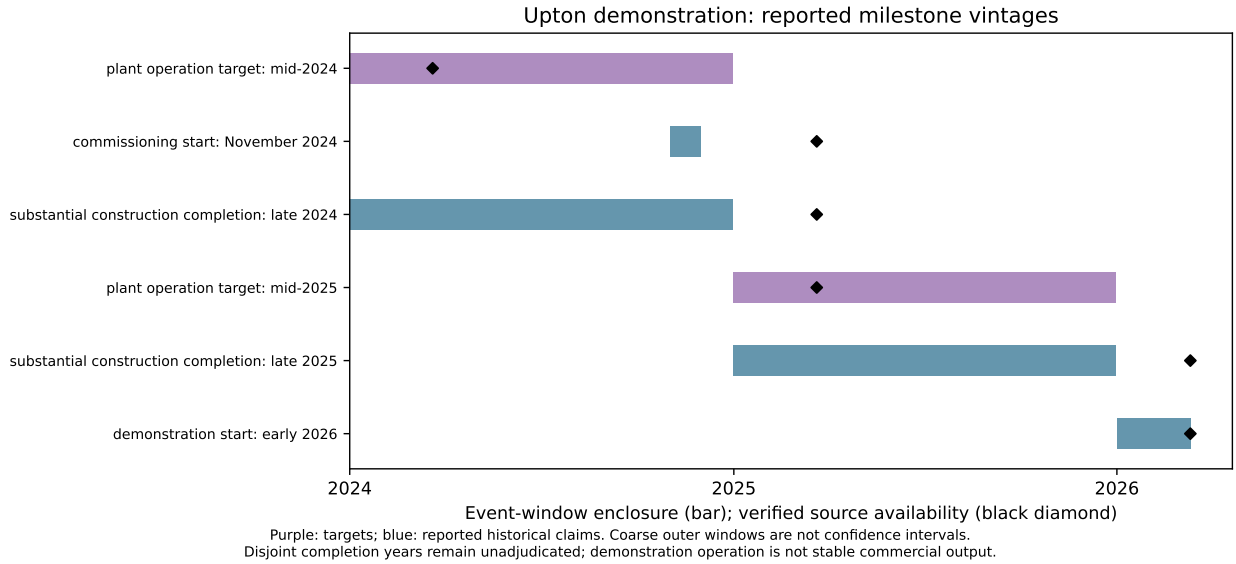


Figure 14: Actual issuer-reported milestone and target versions. Bars preserve coarse event-window enclosures; diamonds give conservative verified source-public dates. No construction-to-operation duration or statistical uncertainty distribution is inferred from these intervals.

upstream feed or unrecorded replenishment is introduced. A second lexicographic objective selects a low-transport allocation within primary-objective tolerance.

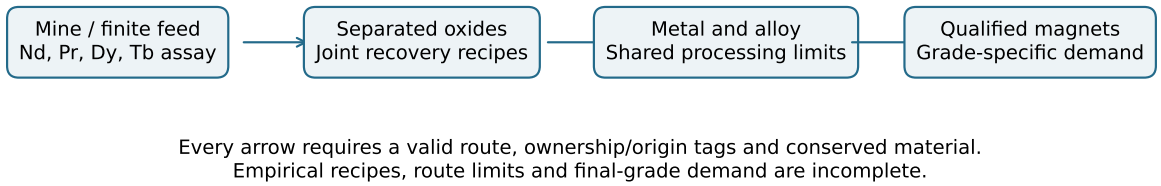


Figure 15: Required physical structure. This is a model schematic, not a solved empirical material-flow diagram.

China exclusion acts on origin/control tags at every relevant node, commodity, inventory and edge. N–1 then removes one additional supplier consistently across its full chain while holding demand fixed. The blueprint's synthetic A/B case reproduces served shares 1, 1/2 and 4/7, with N–1 service 1/2 and uncapped potential 75/70. Four-element and finite-stock tests verify that missing upstream material propagates downstream and that opening inventories cannot be spent every quarter. These are engineering validation results, not estimates of the real importing bloc's resilience.

A source-checked coverage audit supplies the denominator within this selected roster: 35 of 36 activities have a verified source-public clock, and 9 have a reported native annual capacity. One finite demonstration limit is explicitly excluded from annual capacity, despite containing the words "not per year." No complete element recipe, finite route capacity, independently verified stable

qualified rate or matched final-grade demand is verified in this archive. Zero documentation counts indicate missing evidence, rather than zero industrial capability or deliberate opacity.

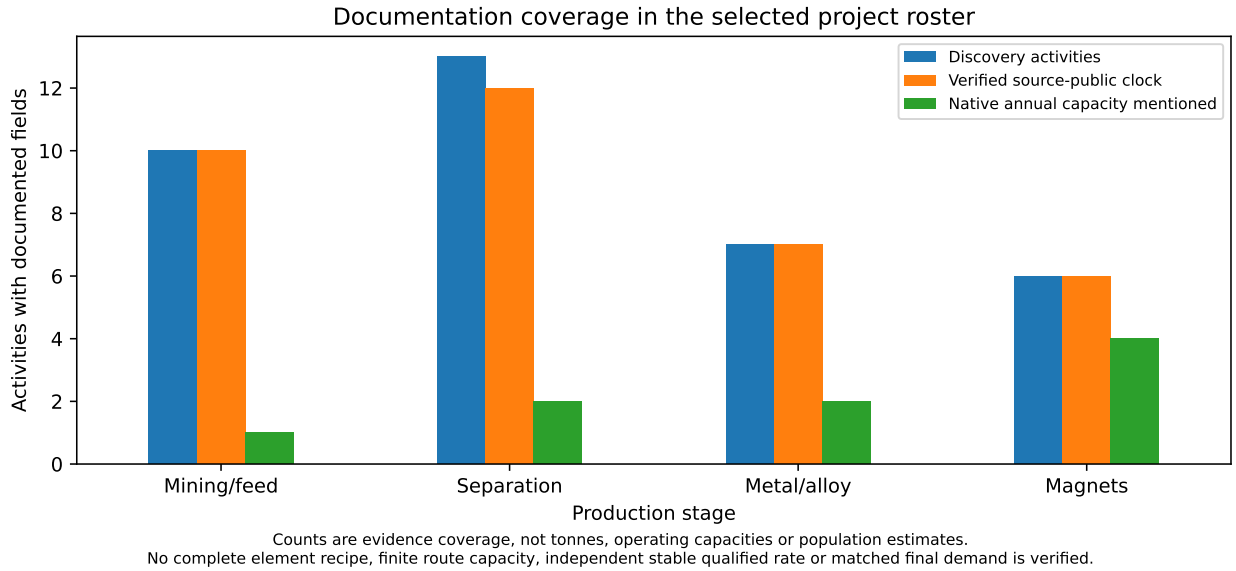


Figure 16: Documentation coverage by stage within the selected discovery roster. Counts are neither tonnes nor a population estimate, and a mentioned annual nameplate may belong to a planned facility.

6.13 Forecasts and historical evaluation

The competing-risk implementation uses $\Lambda_a = \sum_{b \neq a} \lambda_{ab}$,

$$p_{aa} = e^{-\Lambda_a}, \qquad p_{ab} = (1 - e^{-\Lambda_a})\lambda_{ab}/\Lambda_a. \tag{8}$$

Exact, right-censored and interval-censored first-exit observation likelihoods are implemented under stated observation assumptions. The current fitted estimator is a pooled constant-rate benchmark; the fully hierarchical calendar/duration model and a reporting-delay observation process are not empirically estimated.

The synthetic pipeline executes 10,000 independent paths with correlated within-path project shocks and network reoptimization. A separate 30,000-observation synthetic recovery experiment estimates intensities 0.252025 and 0.081212 when the generating values are 0.25 and 0.08. Held-out synthetic Brier score is 0.170032 versus 0.25 for an uninformative half-probability forecast. These figures validate software behavior under known artificial assumptions; they establish no historical forecasting skill.

Requested probabilities for 2028Q4, 2031Q4 and 2035Q4 at 50%, 75%, 90% and 100% service remain missing. A valid empirical network, demand scenario and project risk set are prerequisites. Four-quarter persistence distinguishes the first qualifying quarter from recognition three quarters later. Source-vintage and future-endpoint guards prohibit evaluating unobserved 2026 year-end outcomes at the October freeze.

Two additional retrospective mine-output baseline exercises at 2023 and 2024 year-end origins use only published MP annual output available then. They predict 2024/2025 output from a last

observed year two years before the target, because of disclosure lags. Linear trend has descriptive mean absolute error 3,944.77 native REO tonnes, compared with 6,045.50 for persistence and 6,882.83 for the historical mean. With only two origins at one mine, no statistical skill or population completion claim is supported. This exercise is separate from the blueprint's system-level service backtests.

An additional quarterly exercise uses eight common origins from June 2024 through March 2026 and only Lynas front-page KPI vintages publicly available at each origin. Targets are the following calendar quarter, so publication lags are retained. NdPr product-family mean absolute errors are 292.98 native tonnes for the expanding available-history mean, 355.63 for latest-available persistence and 310.13 for the previous-year seasonal value. For total REO they are 637.27, 928.63 and 961.63 tonnes. No fitted structural model is tested here. Eight dependent origins at one issuer, incomplete contemporary information and changing operating conditions preclude a statistical skill claim or inference about magnet-service probabilities.

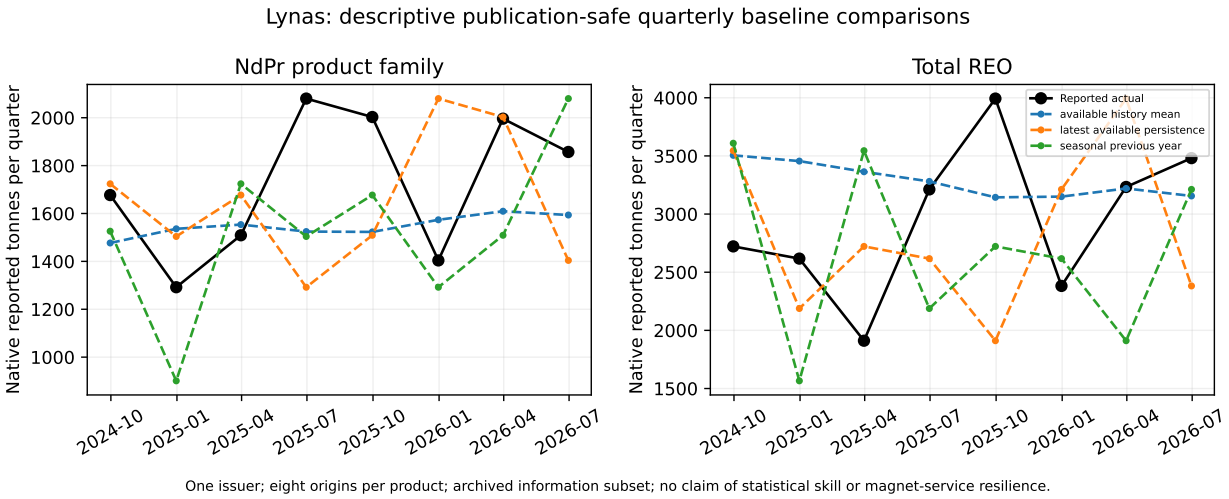


Figure 17: Actual quarterly native-product outcomes and publication-safe naive forecasts. Descriptive eight-origin comparison; no confidence intervals, causal identification or system resilience inference.

The MP quarterly extension evaluates three naive methods on common numerical support within each metric. Its 9 eligible origin–metric pairs span only three quarter-end origins in 2024: three for each upstream KPI, two for NdPr production and one for NdPr sales. Early source N/A/dash cells are not invented seasonal observations. Table 3 reports actual descriptive mean absolute errors, using the latest archived comparative outcomes visible by the freeze. Inputs are separately masked at each origin; changing later comparatives cannot change an earlier forecast. This archive-based exercise does not establish first-release accuracy, statistical superiority or physical-service forecast skill.

7 Semiconductor-control results

The original-rule chronology yields 27 reviewed legal-event phases with publication, effect, jurisdiction, classification and licensing fields. The corpus includes adjustments and authorizations as well as additional restrictions. It is deliberately not transformed into a physical percentage loss or treated

KPI	Origins	Persistence	Seasonal	History mean
REO production	3	2293.00	2325.33	1865.64
REO sales	3	1506.33	1871.00	2392.42
NdPr production	2	295.50	345.50	345.50
NdPr sales	1	458.00	458.00	458.00

Table 3: MP quarterly naive-baseline MAE in each KPI's native metric tonnes. Counts are one to three dependent origins per metric; methods share the same within-metric support. These are descriptive comparisons, not a hypothesis test or a system-service evaluation.

as a complete current-law consolidation.

The additional legal review codes specified YMTC-related Entity List entries effective 16 December 2022 and 13 advanced-computing entities, including Biren and Moore Thread, effective 17 October 2023 (Bureau of Industry and Security, 2022b, 2023b). Their license requirements and denial presumptions are preserved rather than coded as categorical prohibitions. A correction to the 3A090.a license-table entry took effect on 11 February 2025, with public filing that day and Federal Register publication on 14 February (Bureau of Industry and Security, 2025a). July 2026 UAE amendments distinguish specified approved recipients from transactions for which licenses remain required; they are not a blanket China-access relaxation (Bureau of Industry and Security, 2026b).

Forty-one additional original texts received a source-checked scope screen. Eight procedural notices do not themselves add an export requirement; four documents supply the five newly reviewed phases. The remaining entity, general-item, other-destination and import/domestic-policy records require further mapping or remain contextual evidence. In particular, the December 2023 document with "updates and corrections" in its title only extends a comment deadline. Source-public dates, exact operative language and separate compliance dates matter more than title keywords. This single-researcher screen is not an independent legal audit or a complete consolidated law.

A further original-provision extract retains 41 selected claims, 65 operative XML elements and 24 structured technical bounds from documents 2023-23055, 2024-28270 and 2026-00789 (Bureau of Industry and Security, 2023a, 2024a, 2026a). Four parameter-restatement links compare the selected 3A090.a/b numerical branches without asserting full-instrument supersession or unchanged complete scope. The extract uses conservative Federal Register publication clocks and preserves earlier legal effect and later compliance separately. In December 2024, publication was 5 December, effect was 2 December, and selected classification/license instructions had 31 December compliance; the selected HBM-exception instruction followed default timing. Legal effect never substitutes for the information clock.

The selected HBM classification uses bandwidth density strictly greater than 2 GB/s/mm^2, while the separately conditioned HBM exception requires density strictly below 3.3 GB/s/mm^2. The applicable package/stack-area denominator differs from a logic-die performance-density measure. Passing both numerical screens does not establish transaction eligibility: headquarters, routing, end use, documentation and other requirements remain essential. The January 2026 case-by-case review paragraph concerns specified U.S. exports to China/Macau with TPP strictly below 21,000 and DRAM bandwidth strictly below 6,500 GB/s. All nine certification conditions, including the stated aggregate-TPP comparison, are preserved. Neither a review pathway nor a license requirement establishes approval, usable foreign access or successful domestic substitution. Intervening amendments, all underlying jurisdiction/end-use rules, current-law consolidation and independent

specialist adjudication remain incomplete.

7.1 Party-specific schedules and documentary clocks

The additional scope screen identifies 9 archived Entity List rules whose party-level schedules had not been extracted. Their complete tables yield 488 destination-specific entry versions, including 8 address entries, across 27 printed destination headings (Bureau of Industry and Security, 2023c,d,e, 2024b,c,d, 2025b,c,d). Here a printed entry version means a license-bearing block in the archived XML, with all associated entity-cell continuations. Aliases, addresses and repeated country entries do not identify separate firms. Of these versions, 284 occur under the source's China heading and 204 under other headings. This is a selected rule corpus spanning multiple sectors and destinations, not the semiconductor firm population or a complete history of Chinese end-user controls.

The five-column extraction retains exact item scope and review text. Across all destinations, 156 versions state a presumption of denial, 186 a policy of denial, 126 combine denial language with specified case-by-case review, one states case-by-case review alone, and 19 refer to review provisions outside the printed cell. For example, the March 2023 Sichuan University entry states "Case-by-case basis"; the March 2025 rule gives two of its twelve entries a presumption of denial and ten a policy of denial. These distinctions neither establish actual approvals nor imply a numerical access loss. Conditional branches involving food/medicine, telecommunications or other specified items remain attached to their exact source text. The 111 versions referencing notes not reprinted in their particular rule do not imply that the underlying legal provisions or exemptions are absent.

Five rules' unambiguous addition instructions support 248 added entry versions. A 240-record crosswalk binds entries in the four mixed documents to exact named amendatory paragraphs and destination contexts. Together they support 395 additions and 92 revisions; 1 addition candidate remains provisional because the instruction and table spell its name differently. The table's "Mircroelectronics" and instruction's "Microelectronics" are preserved rather than silently corrected or merged. All 150 amendatory instructions and the explicit adjudication notes are retained rather than assigning every table row a new-treatment date. Unique company resolution, historical ownership, transaction scope, intervening amendments and independent legal review remain missing. None of these rows receives a foreign-access, domestic-capability or causal-effect score.

Issued rule	Entries	China heading	Mixed review	Cross-reference
2023-04558	48	38	9	15
2023-07840	34	13	29	0
2023-22536	52	42	0	0
2024-10485	37	37	0	0
2024-19130	131	42	63	0
2024-25411	98	53	16	0
2025-00480	16	14	0	0
2025-05427	12	11	0	0
2025-17893	60	34	9	4

Table 4: Complete entry blocks within the nine selected original XML schedules. Counts include aliases within blocks, address entries and revised versions; they are not counts of newly restricted firms. Mixed review and cross-reference columns describe the source cells across all printed destinations.

The information clock uses the original document's reported official filing date under an end-of-day

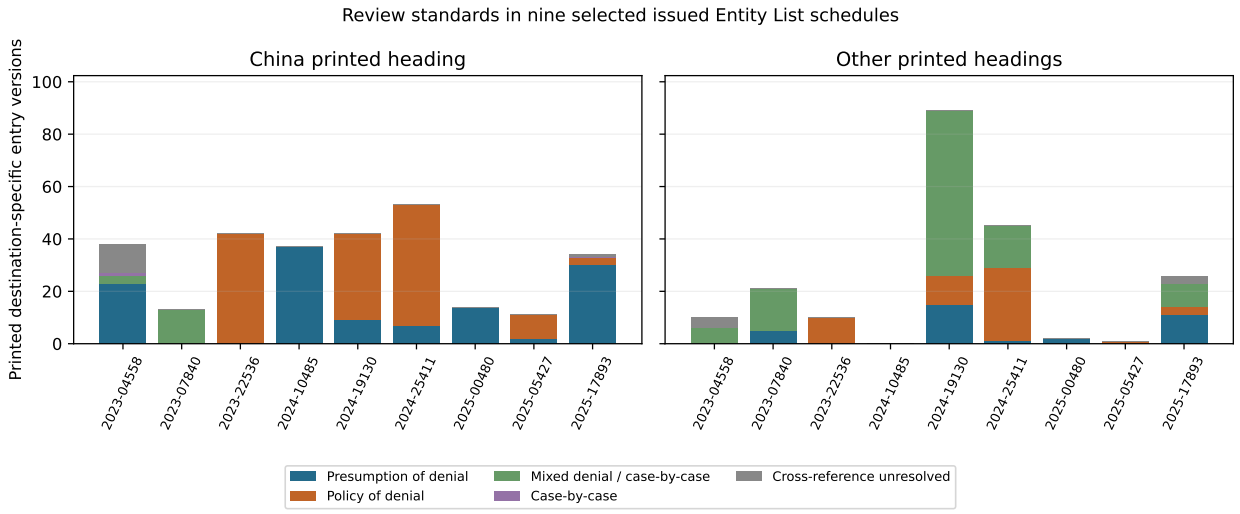


Figure 18: Exact review-language categories, separately for the printed China heading and other headings. Counts describe selected source versions, including revisions, rather than legal intensity, approval probabilities or technology access. The other-heading panel includes rules associated with other destinations and sanctions contexts.

convention; Federal Register publication and legal effect remain separate. Three rules report filing before effect; six were published after effect. A publication-only clock would omit 243 entry versions at their effective dates relative to the reported filing datelines. This is a documentary-clock comparison, not evidence of anticipatory purchasing or independently certified release-time web bytes. The full filing stamp, including its printed time, is retained without inventing a time zone or a globally earliest disclosure date.

Source-format limitations remain material. All 488 citation cells contain unfilled insertion markers, while one address contains an embedded rule-date token. The original XML strings are preserved; metadata citations support bibliographic references without silently repairing the cells. Requests for an earlier public-inspection PDF and the paginated published original failed at the proxy. Until those originals and amendments are checked, this extract certifies the archived XML's contents rather than a clean transcription of the final paginated law. These flags are neither deliberate information withholding nor observations for the opacity index.

7.2 Facility-specific VEU permissions and later removal

The supplied paginated 2024–25411 published PDF corroborates 98 insertion-marker occurrences across fourteen pages and the embedded "EaseyNovember 1, 2024 Commercial Building" address-date text (Bureau of Industry and Security, 2024d). Thus these selected anomalies also occur in the published PDF; they cannot be attributed solely to XML extraction. Their material legal significance is not adjudicated here, and the original text remains unchanged. The supplied 2025–05427 public-inspection version explicitly reports filing at 4:30 p.m. on 25 March 2025 and publication on 28 March; its first page specifies effectiveness on 25 March (Bureau of Industry and Security, 2025c). Reported filing, publication and effect are different clocks, without an asserted globally earliest public timestamp, observed denial, or automatic opacity score.

Two selected original instruments yield 18 source-bound legal passages: 13 operative, one explicitly

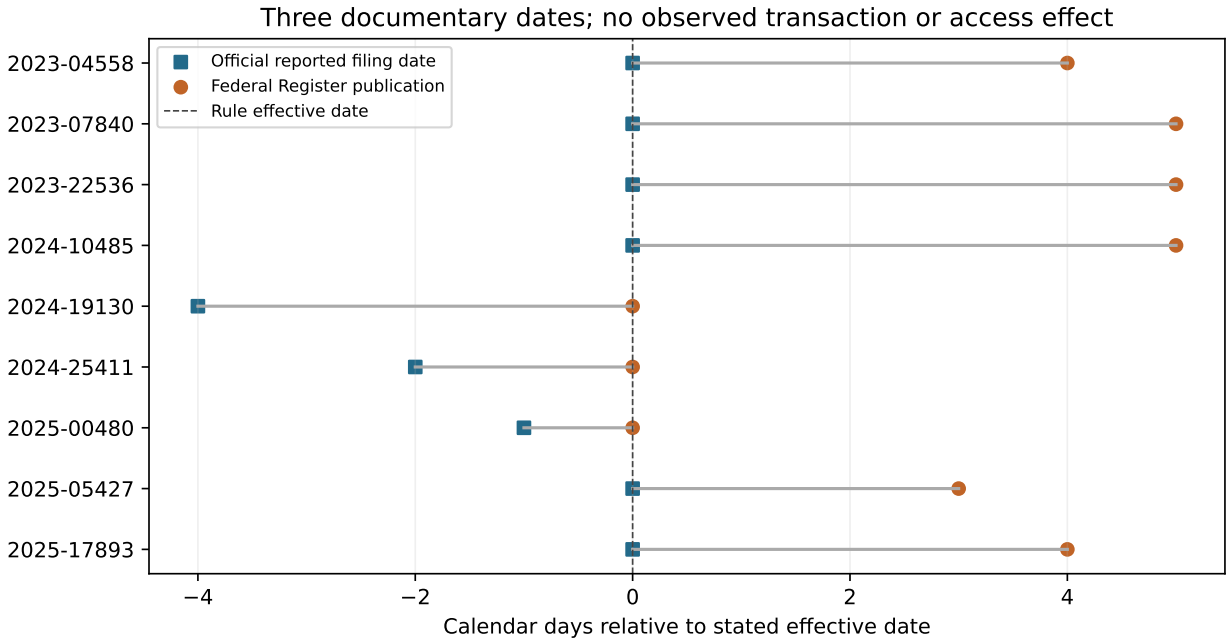


Figure 19: Reported filing and Federal Register publication dates relative to the stated effective date. Negative filing offsets establish documentary notice before effect under this source clock; they do not measure actual trade, anticipation or access.

explanatory and four effective/filing-date passages (Bureau of Industry and Security, 2023f, 2025e). The 2023 rule revises printed permission entries for Samsung's Xi'an NAND facility and SK hynix's Wuxi DRAM facility. Both expressly exclude EUV equipment and the specified related items while retaining the printed targeted exclusions from selected provisions of sections 744.6 and 744.23, the section 748.15(d) civil-end-use condition and the table's non-supersession proviso. This is neither universal permission for all equipment at all Chinese plants nor a numerical access score.

The later rule expressly removes three named recipients—Samsung China Semiconductor, SK hynix Semiconductor (China), and Intel Semiconductor (Dalian)—effective 31 December 2025. Its original reported filing is 29 August and Federal Register publication is 2 September, giving 124 and 120 calendar days respectively before effect. Historical-origin checks preserve the announced future-effective period; knowledge of a removal does not imply that its effect had already begun. The two exact named-recipient links connect selected original instruments, without certifying uninterrupted authorization between them. Intel's prior item/facility scope is not extracted and is left unknown. Removal of a general authorization requires analysis of individual licensing and other applicable conditions; it is not evidence of an observed rejected license or export prohibition.

Named recipient / location	Memory	Scope amendment	Removal
Samsung / Xi'an	NAND	2023-10-17	2025-12-31
SK hynix / Wuxi	DRAM	2023-10-17	2025-12-31
Intel / Dalian	Not extracted	Not extracted	2025-12-31

Table 5: Selected issued facility/item permissions and later explicit removals. This is a documentary comparison, not a consolidated active-authorization or technology-access series.

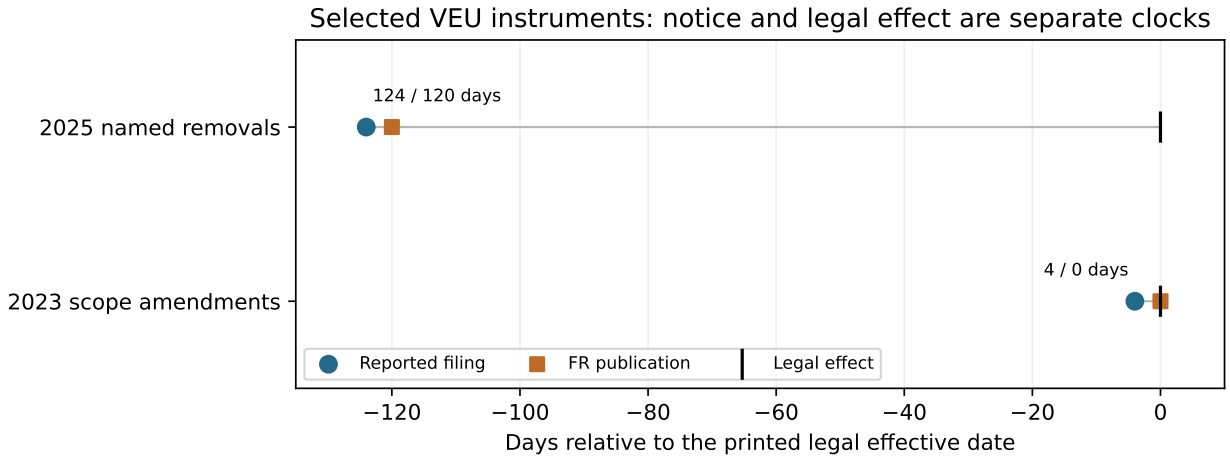


Figure 20: Reported filing, Federal Register publication and legal effect in the two selected VEU instruments. Anticipation and legal effect remain separate.

The original XML citation placeholders remain uncorrected. Two attempted historical statutory-XML retrievals failed, so the earlier 111 unprinted Entity List note references and 19 unresolved review cells remain unresolved. These additional instruments do not establish complete current law, actual approvals, delivered equipment or Chinese domestic functional capability.

7.3 Investment inputs

SMIC earnings releases report nominal annual capital expenditure of USD 6.35, 7.47, 7.33 and 8.10 billion for 2022–2025. Fourth-quarter R&D expenses are USD 197.455, 189.138, 217.035 and 239.676 million, respectively (Semiconductor Manufacturing International Corporation, 2026a); earlier releases and exact locators are in the source manifest. These are inputs with distinct measurement windows. They do not show whether policy caused spending, whether useful domestic substitution occurred, or whether spending would have been higher without restrictions.

7.4 Operating scale, reporting definitions and nominal revenue

Four original releases add 120 versioned operating/input observations, 12 source definitions and four first-page issuer datelines (Semiconductor Manufacturing International Corporation, 2023, 2024, 2025, 2026a). They cover 9 selected quarterly periods, not a continuous panel. Twenty-four comparative groups repeat earlier numerical values; one shipment-basis label and one utilization-denominator wording differ across reporting vintages. Generic 8-inch equivalents and standard-logic 8-inch equivalents therefore remain versioned source descriptions, without an invented conversion or adoption date. Annual management summaries remain separate from quarter-level measures.

For Q4 2025 SMIC reports 2,514,970 standard-logic 8-inch equivalent shipments, reported monthly capacity of 1,058,750 equivalents and quarterly utilization of 95.7%. Its annual/annualized utilization summary is 93.5%, a different reported window. The detailed monthly-rate sentence does not establish an exact measurement day; the dataset endpoint indexes its reporting quarter. The utilization note uses wafers out divided by estimated quarterly capacity; shipments are a different object, and reported monthly capacity does not identify the quarter-average denominator. Multiplying the

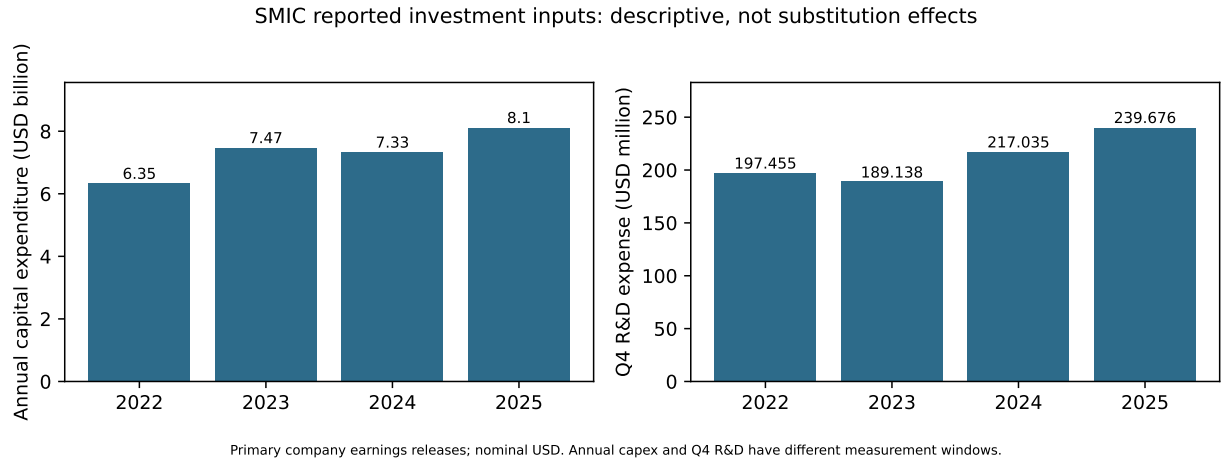


Figure 21: Actual company-reported investment inputs in nominal USD. Annual capital expenditure and fourth-quarter R&D are separate series, not comparable annual totals or causal treatment effects.

reported monthly rate by three and utilization does not reconstruct an issuer production identity. Similarly, the reported 77.2% twelve-inch share is a share of wafer revenue, not physical wafer output or advanced-node production.

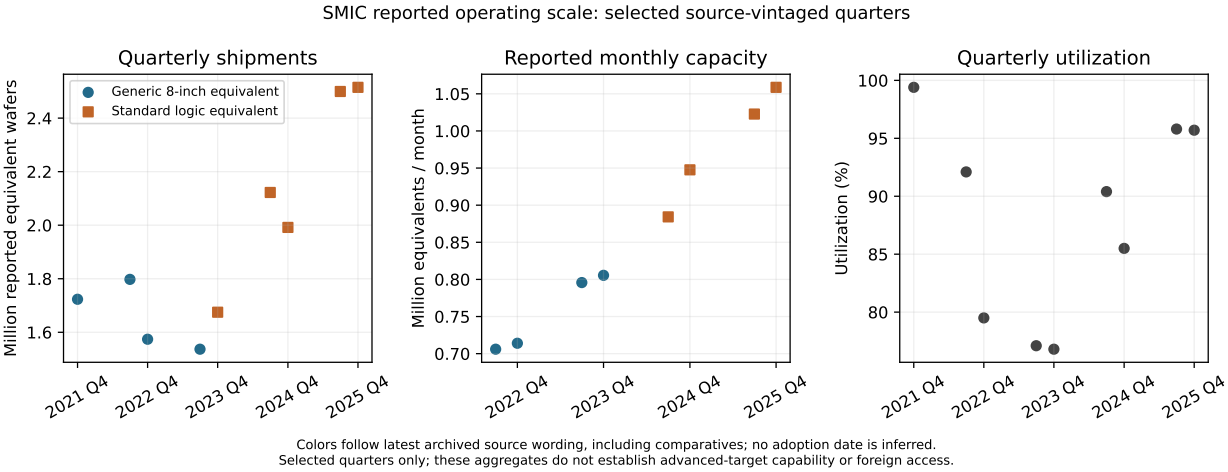


Figure 22: Issuer-reported operating scale for selected quarterly periods. Colors follow the selected latest-by-freeze source's unit wording, including retrospective comparatives; they do not date a technological or reporting-method transition. Missing quarters are not interpolated. Company aggregates do not identify indigenous tools, target-node yield or usable foreign access.

Let R_t be reported nominal total revenue, s_t the reported wafer-service revenue share and Q_t reported equivalent shipments. Define a reconstructed wafer-attributed revenue $R_t^w = s_t R_t$ and a descriptive unit revenue ratio $v_t^{\text{rep}} = R_t^w / Q_t$. This construction implies

$$\Delta \log R_t^w = \Delta \log Q_t + \Delta \log v_t^{\text{rep}}. \tag{9}$$

The code verifies 8 within-source quarter-on-quarter/year-on-year accounting decompositions. Because shares are rounded and products, equivalent-wafer conventions and revenue recognition can

differ, v_t^{rep} is not a quality-adjusted price. Its change cannot isolate pricing, product mix, technical performance or a control effect.

In the latest annual release's Q4 2025 versus Q4 2024 comparison, reported shipments increase by approximately 26.3%, reconstructed wafer-attributed nominal revenue by 12.6%, and the unit revenue ratio changes by -10.8%. Table 7 retains all four selected source-year comparisons. These figures establish descriptive scale/revenue accounting; they do not identify advanced-technology substitution or a causal policy response. One consolidated firm, sparse selected quarters, unknown target yields/tool origins, overlapping restrictions and no defensible untreated comparator prevent the requested access/adaptation estimation.

The supplied Q1 and Q2 2026 releases add 56 native cell versions with issuer release datelines of 14 May and 13 August (Semiconductor Manufacturing International Corporation, 2026b,c). Reported standard-logic eight-inch-equivalent shipments are 2,509,137 and 2,869,495; monthly capacity rates are 1,078,250 and 1,096,500; utilization is 93.1% and 93.7%. Four new within-source accounting decompositions reconcile. For Q2 year-on-year, shipments rise 20.1%, reconstructed nominal wafer revenue 36.1%, and revenue per reported equivalent 13.3%. Forty-four repeated cell groups across old and supplied releases agree numerically; definition wording remains source-specific. None of these aggregates identifies advanced-node yields, indigenous tool origin or remaining usable foreign access.

The same pair supports a narrow historical forecast check. On 14 May, already within Q2, SMIC guided Q2 revenue growth of 14–16% quarter-on-quarter and gross margin of 20–22%. The later Q2 release reports approximately 20.0% total revenue growth and 25.3% margin, above both upper ranges. Translating the revenue guidance using the origin's exact native revenue table gives a midpoint error of approximately minus USD 124.278 million, versus minus USD 500.101 million for persistence. Margin midpoint error is minus 4.3 percentage points, versus minus 5.2 for persistence. The midpoint is our evaluation convention, not an issuer point forecast; the stated ranges are not probabilistic intervals. Prior-year Q2 numbers first observed in the selected archive through the later release are not backdated into a seasonal baseline. Two correlated financial outcomes in one target quarter cannot establish general forecasting skill, validate the structural research model, or substitute for the prescribed longer-horizon capability forecasts.

SMIC comparison	Shipments	Wafer revenue	Revenue/equivalent
2026Q1 QoQ	-0.2	2.3	2.5
2026Q1 YoY	9.5	10.0	0.5
2026Q2 QoQ	14.4	20.9	5.7
2026Q2 YoY	20.1	36.1	13.3

Table 6: Additional source-native SMIC percentage changes. Rounded revenue shares enter reconstructed wafer revenue; the ratio term is neither a quality-adjusted price nor a causal adaptation effect.

7.5 Target capability and foreign access remain distinct

The measurement implementation separates domestic layer capability z_ℓ, foreign usable access f_ℓ, and effective availability $a_\ell = z_\ell + (1 - z_\ell)f_\ell$. The latter is a residual-gap approximation; a complementary-input sensitivity uses a different production rule. Strict minima are retained across essential layers. Unobserved layers propagate missingness or logical bounds, not midpoint or zero

Q4 vs. prior Q4	Shipments	Wafer revenue	Revenue/equivalent
2022	-8.7	4.4	14.3
2023	6.4	5.6	-0.8
2024	18.9	31.0	10.1
2025	26.3	12.6	-10.8

Table 7: Source-native Q4 year-on-year percentage changes reconstructed from reported quantities and rounded shares. These are descriptive accounting comparisons, not causal estimates or quality-adjusted prices.

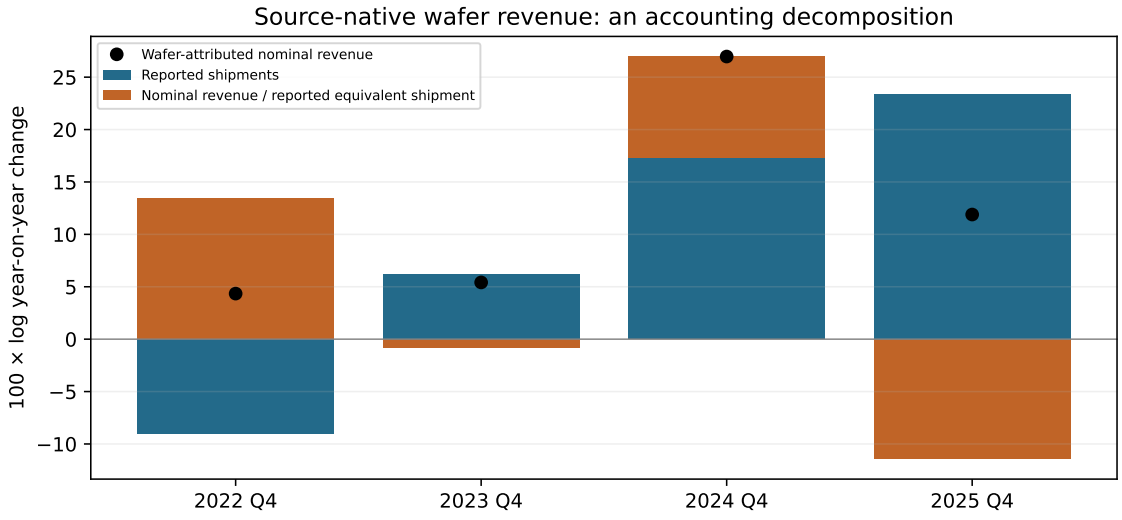


Figure 23: Exact decomposition of the constructed nominal wafer revenue measure, in $100 \times \Delta \log$ units. Positive and negative contributions are displayed around zero; black points mark the net change. Table 7 uses ordinary percentage changes. No contribution is interpreted as technological adaptation or a causal restriction effect.

imputation.

Two provisional target alternatives specify a compute-plus-HBM stack and a logic-only stack, with explicit throughput, yield, power and memory requirements. They are researcher sensitivity specifications, not established engineering standards, and lack specialist approval. No country capability scores are assigned. Consequently neither the slow domestic game state x nor a causal access/adaptation elasticity is empirically available. Aggregate imports, patents, spending and a nominal node label cannot fill this gap.

8 Strategic opacity results

The ten-indicator SPOI codebook is implemented with disclosure, predictability and clarity weights 0.40/0.35/0.25. Mandatory disclosure lateness/completeness, licensing-statistics availability, policy lead times, scope clarity, adverse reversals, official contradictions, missing obligatory releases, stage-classifiable output and amendment counts have distinct observable denominators. Missing research data and inapplicable duties are separately coded.

For observed indicators,

$$\mathrm{SPOI}_d = 100(0.40D_d + 0.35P_d + 0.25L_d), \tag{10}$$

where each pillar is its mean indicator score divided by four. At least eight indicators, including three disclosure, two predictability and one clarity indicator, must be observed. Quarter means require valid observations on at least 80% of calendar days. Defined no-event rules return zero only under verified event-search coverage; they do not imply that every unobserved policy is transparent.

The specified synthetic vectors reproduce 57.416667 and 16.75 exactly within numerical tolerance. Dated-obligation, reversal-window, legal-effect, amendment-deduplication, missingness, reliability and quarter-coverage tests pass. In contrast, the current primary China-policy corpus and required disclosure denominators do not support a coverage-valid historical headline index. There are zero valid historical quarters in the executed coverage audit. This is a statement about evidence coverage, not measured high or low Chinese opacity.

There is no independently blinded second-coder historical dataset. Reliability software and synthetic agreement checks cannot replace it. No opacity-delay or quadratic overreaction coefficient is estimated. In particular, an apparent turning point $-\beta_1/(2\beta_2)$ would require identified curvature, observed support and uncertainty analysis before interpretation.

9 Reciprocal dynamic game

9.1 States, policies and payoffs

States are Chinese domestic semiconductor substitutability x, bloc China-free final-magnet service y, and delayed policy-exposure pipelines m, n. Controls $u, c \in [0, 1]$ represent normalized U.S.-led semiconductor and Chinese mineral restrictions. Game time is years; a quarter is 0.25 years. The coherent bounded law is

$$\dot{m} = \kappa_C(u - m), \qquad \dot{n} = \kappa_U(c - n), \tag{11}$$

$$\dot{x} = g_C e^{\beta_C m}(1 - x) - \delta_C x, \qquad \dot{y} = g_U e^{\beta_U n}(1 - y) - \delta_U y. \tag{12}$$

Positive g_i, κ_i, nonnegative depreciation and sign-unrestricted β_i make the state hypercube forward invariant. Fixed controls imply $m^* = u, n^* = c$ and

$$x^*(u) = \frac{g_C e^{\beta_C u}}{g_C e^{\beta_C u} + \delta_C}, \qquad y^*(c) = \frac{g_U e^{\beta_U c}}{g_U e^{\beta_U c} + \delta_U}. \tag{13}$$

The constant-policy Jacobian has negative eigenvalues $-\kappa_C, -\kappa_U, -(g_C e^{\beta_C u} + \delta_C), -(g_U e^{\beta_U c} + \delta_U)$. These do not establish closed-loop stability.

Discounted objectives use $\rho_i > 0$ and flow payoffs

$$\pi_U = A_U u(1-x) - D_U c(1-y) - \tfrac{1}{2}k_U u^2 - \tfrac{1}{2}h_U n^2 + B_U y + q_U uc, \tag{14}$$

$$\pi_C = A_C c(1-y) - D_C u(1-x) - \tfrac{1}{2}k_C c^2 - \tfrac{1}{2}h_C m^2 + B_C x + q_C uc. \tag{15}$$

These are normalized hypothetical utility functions. Their coefficients are not estimates of national welfare or strategic preferences. The primary scenario uses $g_C = 0.12, g_U = 0.10$, $\delta_C = 0.08, \delta_U = 0.07$, $\kappa_i = 0.5$, $\rho_i = 0.2$, $A_i = 0.8$, $D_i = 0.5$, $k_i = 1$, $h_i = 0.05$, $B_i = 0.2$, $q_i = 0.15$, and $\beta_i = 0.7$. Negative/zero response and other payoff scenarios are separately registered.

9.2 Constrained actions and the dynamic equilibrium distinction

The stationary HJB equations are $\rho_i V_i = \max_{a_i}\{\pi_i + \nabla V_i \cdot f\}$, holding the opponent's feedback policy fixed. Conditional on continuation gradients,

$$M_U = A_U(1-x) + \kappa_C V_{U,m}, \qquad M_C = A_C(1-y) + \kappa_U V_{C,n}, \tag{16}$$

and simultaneous best responses are

$$u = \mathrm{clip}_{[0,1]}[(M_U + q_U c)/k_U], \qquad c = \mathrm{clip}_{[0,1]}[(M_C + q_C u)/k_C]. \tag{17}$$

All nine lower/interior/upper active regimes are implemented and checked against conditional deviations. For $k_U = k_C = 1, q_U = q_C = 0.8, M_U = 1.2, M_C = 0.1$, the correct pair is $(1, 0.9)$; independently clipping the unconstrained simultaneous interior solution gives the incorrect pair $(1, 1)$. Singular interior systems can admit an equilibrium line segment and require separate handling.

The sparse HJB discretization has inward state-constrained stencils and a monotone Markov-chain generator. Because an action changes the sign of pipeline drift, the discrete Hamiltonian is piecewise quadratic. Its improvement step evaluates both directional pieces and the common kink, rather than using one frozen derivative. This is a documented numerical repair, preserving the continuous model's nine analytical regimes.

9.3 Executed diagnostics

Six initial policies are evaluated for the positive-response case at $17^4 = 83{,}521$ and $21^4 = 194{,}481$ states. Separate sparse equations are solved for both players. Recorded diagnostics include linear/HJB residuals, global one-period continuation-value deviation gains, action errors, archived values/policies, runtimes, grid changes and off-grid residuals. Table 8 is regenerated directly from the archived independent audit.

Positive-response policy sup changes decrease with refinement: approximately 0.00319 from 7 to 17 nodes per dimension and 0.00135 from 17 to 21. Holding the coarser policy on the finer grid gives a

Scenario	Grid	HJB sup	Gain sup	Action error	Status
negative	7	2.60e-11	8.33e-11	8.19e-08	candidate
negative	17	1.75e-10	3.34e-02	3.13e-01	not solved
negative	21	1.29e-10	2.48e-03	7.05e-02	not solved
positive	5	3.49e-11	3.43e-10	8.59e-08	candidate
positive	7	4.48e-11	2.67e-10	9.66e-08	candidate
positive	17	9.69e-11	3.89e-14	9.96e-08	candidate
positive	21	8.46e-11	4.95e-15	9.94e-08	candidate
zero	7	2.51e-11	2.67e-15	7.30e-08	candidate
zero	17	4.69e-11	2.67e-15	7.30e-08	candidate
zero	21	1.26e-10	2.67e-15	7.30e-08	candidate

Table 8: Hypothetical finite-grid diagnostics. For each scenario/mesh the displayed errors are maxima across saved starts or methods. Candidate labels apply only to the finite-state approximation.

unilateral gap of about 7.15×10^{-7} in the latter comparison. Off-grid continuous HJB residuals remain around 0.006 in the 21-node interpolated solution. These discrepancies exceed the finite-grid residuals and preclude a continuum-equilibrium certificate.

Negative-response runs fail fine-grid acceptance under both joint and alternating-response algorithms despite small linear equation residuals. This numerical failure is neither proof of nonexistence nor a policy finding. It demonstrates that solving policy-evaluation equations is insufficient. Closed-loop stationary-point diagnostics include policy derivatives; grid-knot roots are labeled inconclusive when central secants cannot establish a local Jacobian.

9.4 Additional negative-adaptation attempts

Three further initial policies—zero, one and seeded random—were executed at 17^4 states for 160 iterations each, with the same negative-response parameters and unchanged acceptance criteria. All 3 attempts are recorded as not solved (Table 9). These are additional to the earlier three failures, while the earlier 34 accepted finite-grid runs remain unchanged. The maximal unilateral gaps range from 3.20×10^{-5} to 2.48×10^{-2}, exceeding 10^{-8}; action residuals range from 0.0118 to 0.311, exceeding 10^{-7}. Independent saved-array evaluation verifies the reported failures. It does not turn them into equilibrium candidates or establish nonexistence. Figure 24 shows why a small linear solve residual cannot replace the nonlinear checks.

Start	Iterations	Deviation gain	Action residual	Linear residual
zero	160	3.2e-05	0.012	1.1e-10
one	160	0.001	0.077	1.3e-10
random	160	0.025	0.31	1.9e-10

Table 9: Additional hypothetical negative-adaptation solver failures. Each start exhausts 160 iterations; no row satisfies the Nash criteria.

9.5 Separately registered safeguarded search

A different iteration searches for roots of the evaluated global-response residual $r(z) = T(z) - z$, where z stacks both policy arrays and T first evaluates their original discounted values and then

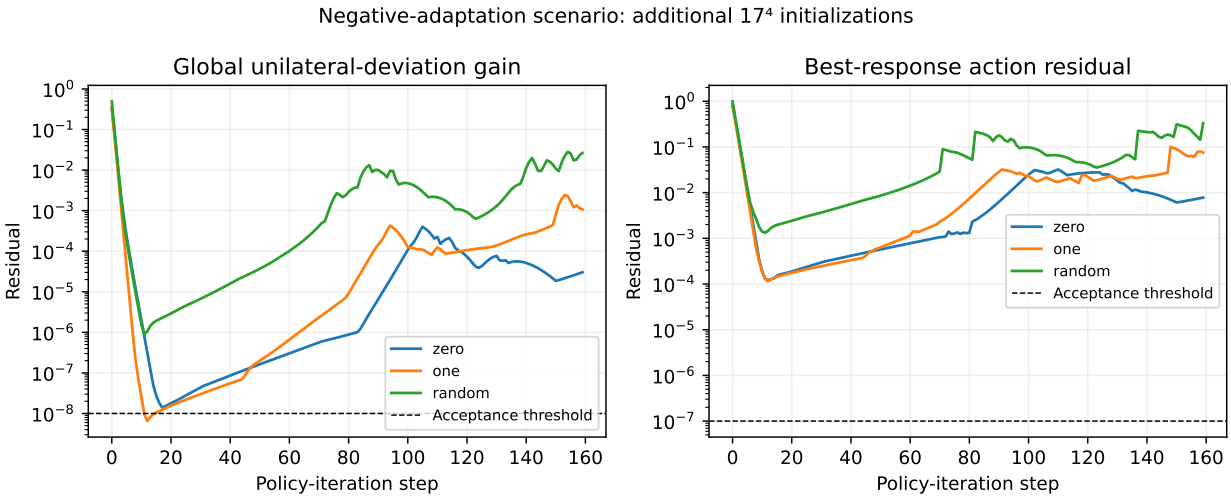


Figure 24: Registered negative-adaptation iterations at 17^4 nodes. Horizontal lines are the unchanged unilateral-gain and action-error thresholds. These are numerical diagnostics, not empirical state probabilities or evidence of nonexistence.

computes the two global unilateral Hamiltonian maximizers. A standard multisecant Anderson construction proposes

$$z_{\text{trial}} = \text{proj}_{[0,1]^{2N}}\{z + \omega r - (\Delta Z + \omega \Delta R)\gamma\}, \tag{18}$$

with regularized least-squares coefficient γ fitted to recent residual differences. Every bounded trial is freshly policy-evaluated; backtracking compares a registered infinity or RMS merit, with a relaxed unaccelerated response direction as fallback. Projection ensures admissibility only. Neither projection nor merit reduction establishes a Nash condition, and no contraction/convergence theorem for this general-sum response map is asserted. The original four-state law, linear solver, exact directional maximizers and all original final acceptance thresholds remain unchanged.

The 12 registered attempts consist of six infinity-merit searches at 7^4 and 17^4 nodes, three RMS-merit searches at 17^4, and three 17^4 searches initialized by interpolation from the verified coarse policies. The three accepted coarse runs are separately checked; they are not three proved distinct equilibria. All nine fine-grid attempts fail the original criteria and stop when no tested residual-decreasing direction is found. For example, the infinity-merit zero start has gain 5.18×10^{-9} but action error 1.02×10^{-4}, which fails 10^{-7}. A smaller gain cannot substitute for the action condition. No further 21-node run is justified as a refinement of a fine candidate from this method because none was accepted.

Search	Grid	Accepted	Attempts	Max gain	Max action error
linf	7^4	3	3	8.3e-12	9.8e-08
linf	17^4	0	3	3.3e-06	0.0026
rms	17^4	0	3	5.1e-06	0.0032
coarse interpolation	17^4	0	3	6.6e-09	0.00011

Table 10: Hypothetical safeguarded-search outcomes. Counts are numerical attempts, separate from the original 34 accepted finite-grid runs and three earlier failures, and from the three additional failures above. They are not empirical observations or distinct-equilibrium counts.

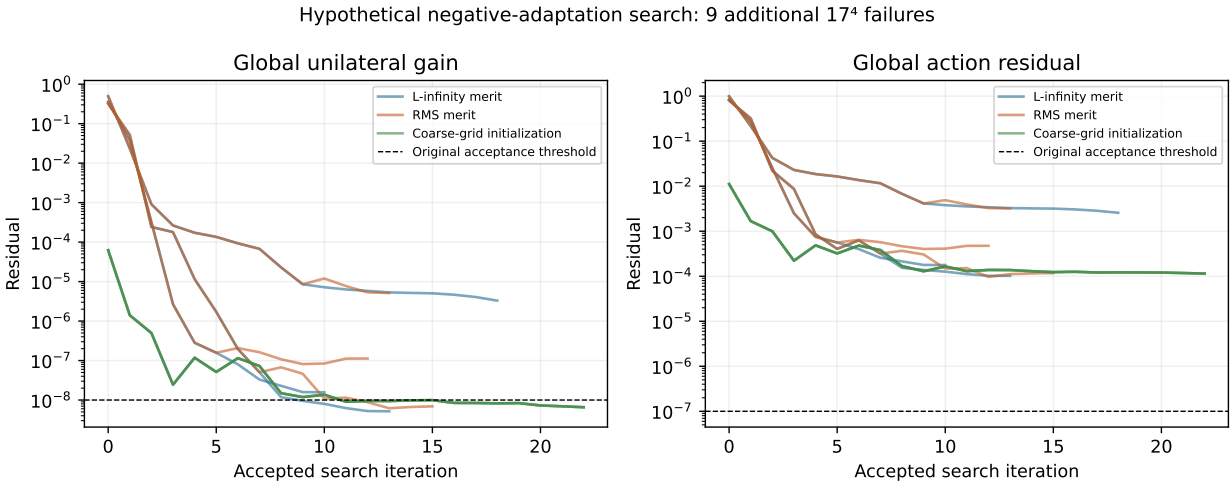


Figure 25: Registered negative-adaptation safeguarded searches. Search merit differs across families; final acceptance always uses the original global gain, action and evaluation conditions. Nine fine-grid failures are retained.

Each final attempt pins the executed numerical source and registration; initial code versions, arrays and logs remain separately archived. An independent original-HJB recheck evaluates all final policies and reconstructs their initial/warm-interpolated policies. These checks establish archive/status integrity and conditional finite-grid diagnostics. They do not establish a continuous pure MPE, nonexistence, empirical calibration or a convergence-speed advantage over algorithms that performed different work.

9.6 Independent discounted unilateral deviations

A separate audit holds each archived opponent feedback policy exactly fixed and solves the other player's discounted finite-state control problem. Four original archives yield 8 unilateral solves: positive-response zero starts at 17 and 21 nodes per dimension, and two previously failed negative-response 17-node policies. All eight one-player solves meet their registered global-Hamiltonian and policy-evaluation conditions, requiring one iteration for the positive cases and three for the negative cases. The original full-game status labels are unchanged.

For the two positive-response archives, the residual-adjusted discounted-regret upper endpoints range from 1.80×10^{-10} to 5.34×10^{-10} across players and meshes. For the failed negative midpoint archive, explicit unilateral feedback policies give residual-adjusted lower gains of 1.3213×10^{-4} for the U.S. player and 1.5782×10^{-4} for China. For the failed additional zero-start archive, the corresponding lower gains are 1.1281×10^{-5} and 1.2850×10^{-5}. Each quantity maximizes over model grid states; the maximizing states can differ between players. They are not observed welfare gains, empirical state probabilities or an aggregate international welfare comparison.

Appendix A.5 derives the comparison bounds. This extends verification from a single-state deviation to discounted unilateral control against fixed opponents in the same finite-state approximation. It does not verify a deterministic continuous-state MPE, remove interpolation errors, solve a different empirical game or establish nonexistence in the failed cases.

Archive	Player	Iterations	Lower gain	Upper regret
positive n17 zero	U.S.	1	0	2.4e-10
positive n17 zero	China	1	0	5.3e-10
positive n21 zero	U.S.	1	0	4.5e-10
positive n21 zero	China	1	0	1.8e-10
negative n17 mid	U.S.	3	0.00013	0.00013
negative n17 mid	China	3	0.00016	0.00016
negative n17 zero	U.S.	3	1.1e-05	1.1e-05
negative n17 zero	China	3	1.3e-05	1.3e-05

Table 11: Hypothetical fixed-opponent discounted-control audit. Bounds use measured numerical residuals and ordinary floating-point calculations, without interval certification. Original full-game labels are preserved.

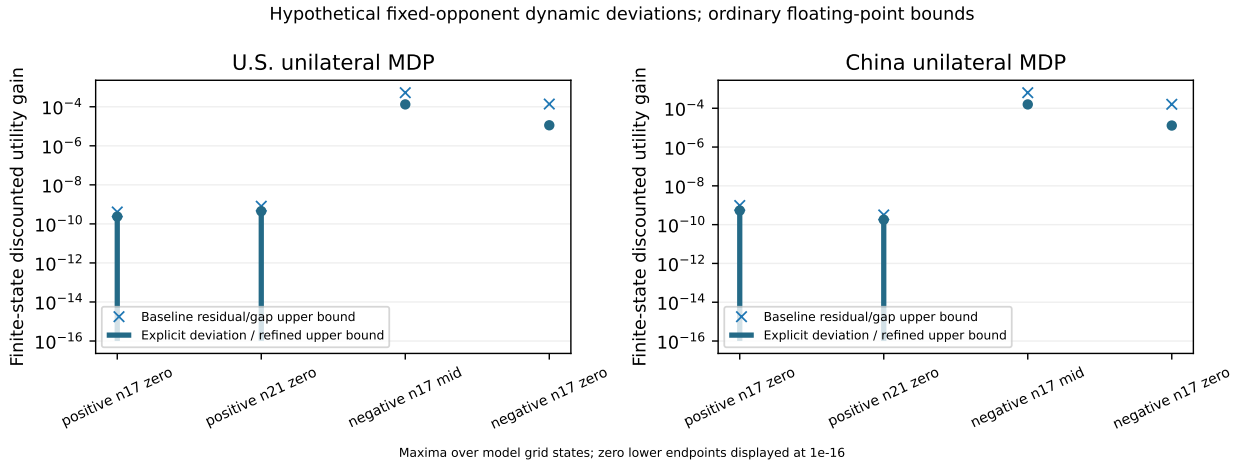


Figure 26: Residual-adjusted discounted-regret diagnostics for finite-state hypothetical policies. Zero lower endpoints remain zero in the data and are displayed at 10^{-16} solely on the logarithmic axis.

9.7 Reciprocal commitment to constant paths

A separate fallback retains the original reciprocal four-state law and full payoffs but restricts each player's time-zero choice to a constant control forever, starting at the hypothetical state $(.25, .25, 0, 0)$. Unilateral deviations are also constants. This restricted commitment Nash is neither a feedback MPE nor an equilibrium against unrestricted time-varying open-loop deviations. Because own-action state subsystems separate, the unilateral objective, up to an opponent-dependent constant, is

$$F_i(v; w) = A_i v W_i(v) - \frac{k_i v^2}{2\rho_i} + \frac{q_i v w}{\rho_i}, \qquad W_i(v) = \int_0^\infty e^{-\rho_i t}[1 - z_i(t; v)]\, dt, \tag{19}$$

where $(z_U, z_C) = (x, y)$. Appendix A.3 derives sufficient strict-concavity and best-response contraction conditions that guarantee a unique Nash in this restricted class for a fixed initial state, including boundary choices. This uses standard concavity and contraction arguments; it does not establish a general dynamic-game existence theorem.

All 9 registered hypothetical scenarios satisfy these sufficient bounds. Across 27 initializations, the largest action spread is 5.23×10^{-12}. The numerical checks include 2322 unilateral profile points, an exact zero-response reduction, 54 independent full-original-payoff comparisons and 18 tighter-quadrature checks. The maximal refined gradient KKT error is 5.75×10^{-12}; the maximal full-payoff decomposition error is 3.51×10^{-12}. The maximal analytic omitted infinite-payoff tail bound is 2.76×10^{-13}. Standard ODE integration is not interval certified; numerical error estimates and exact mathematical sufficient conditions are distinct evidence.

Scenario	u	c	μ_U	$L_U L_C$
adaptation negative	0.6286	0.6382	1.839	0.136
adaptation zero	0.5436	0.5572	5.000	0.022
adaptation positive	0.4691	0.4836	2.385	0.085
patient	0.4015	0.4149	1.756	0.472
impatient	0.5466	0.5597	1.787	0.041
substitution	0.3561	0.3745	2.385	0.085
complementarity	0.6311	0.6436	2.385	0.605
industrial subsidy growth	0.4619	0.4289	2.385	0.151
foreign licensing relaxation	0.3297	0.4652	3.366	0.060

Table 12: Hypothetical reciprocal constant-commitment Nash actions. The contraction bound certifies uniqueness only within the constant-path strategy class at a fixed initial state.

With otherwise canonical scenario parameters, (u, c) falls from approximately $(.62864, .63822)$ at negative response to $(.54358, .55721)$ at zero response and $(.46906, .48359)$ at positive response. This is a conditional comparison, not a theorem about all reciprocal games or an empirical policy recommendation. Numerical horizons of 80–320 years bound discounted mathematical tails; they are not industrial forecasts. These infinite-horizon constant-path payoffs are distinct from the 30-year, no-terminal-continuation feedback-trajectory comparisons below.

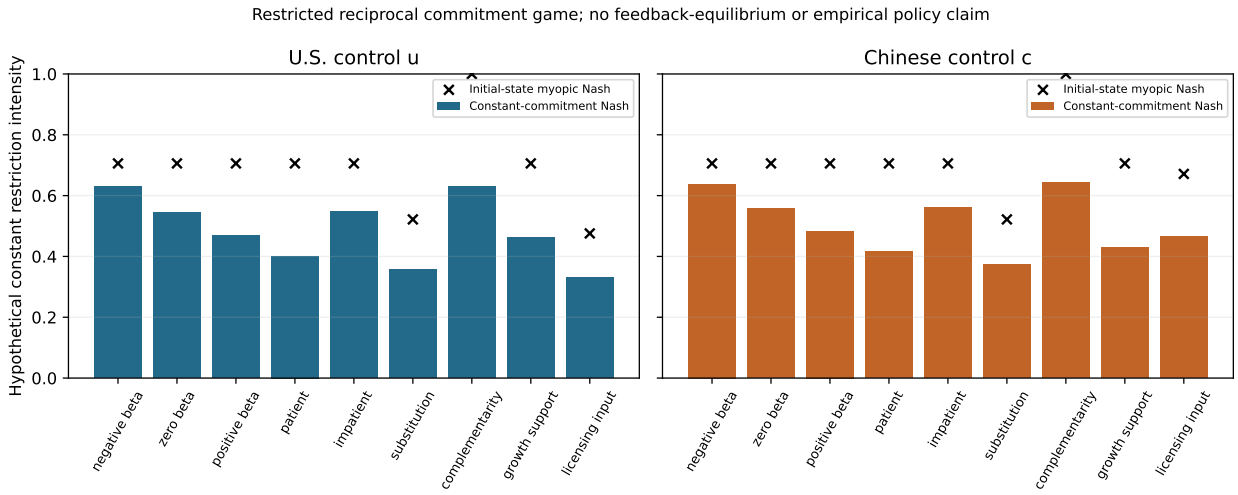


Figure 27: Restricted constant-commitment actions versus the contemporaneous myopic action game at the same hypothetical initial state. All inputs are scenarios; no observed national policy is fitted.

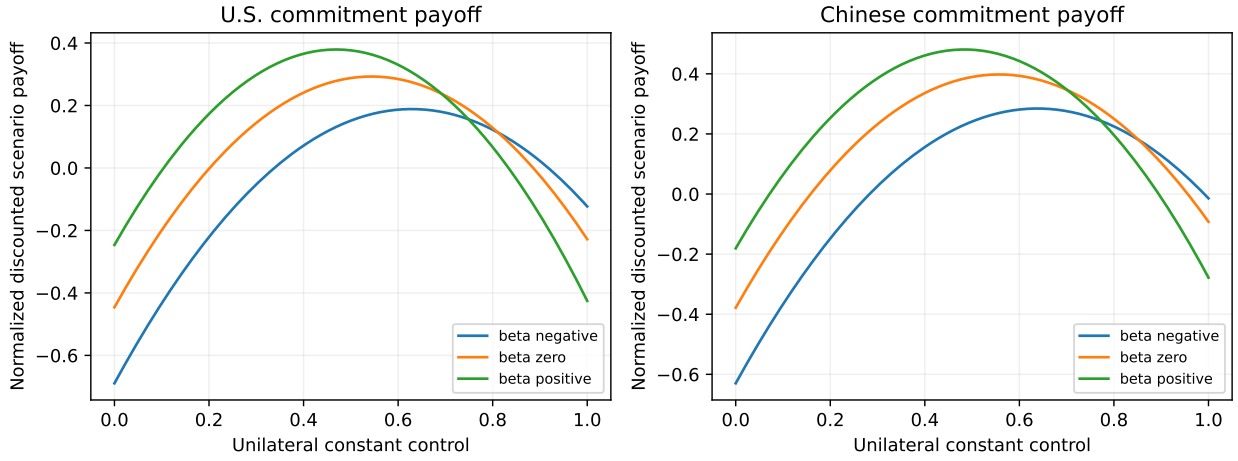


Figure 28: Full original discounted payoffs under unilateral constant deviations holding the opponent at its restricted commitment action, for three adaptation-sign scenarios. Curves use the own-action decomposition, independently checked against full four-state payoff integration. These are arbitrary hypothetical utility units.

10 Counterfactuals and robustness

The one-sided constant-control benchmark supplies a limited analytical result. For $\dot{x} = bu(1-x)$ and vulnerability $v_0 = 1 - x_0$, constant u gives

$$J(u) = \frac{Auv_0}{\rho + bu} - \frac{ku^2}{2\rho}, \qquad ku(\rho + bu)^2 = Av_0\rho^2. \tag{20}$$

The left side increases for $u \geq 0$, so there is a unique nonnegative root. For positive b, the interior forward-looking optimum is below the comparable interior myopic action. With all five positive benchmark parameters equal to one, the constrained optimum is 0.4655712319, versus myopic action one. This does not solve the unrestricted time-varying problem or prove restraint in the reciprocal game.

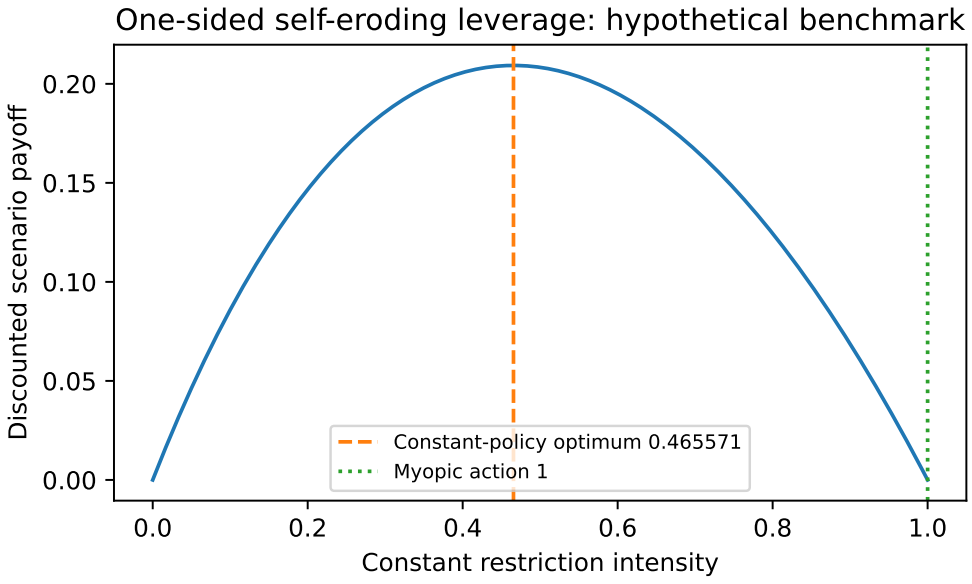


Figure 29: Analytical hypothetical constant-control benchmark. Parameters are test assumptions, not estimated national preferences.

Registered game comparisons vary positive/zero/negative induced adaptation, discounting, complementarity/substitution, baseline growth support and an immediate coercion-value parameter representing licensing relaxation. Trajectories compare candidate feedback, controls scaled 25% up/down, and a unilateral one-year shock. These are theoretical parameter mappings. The observed-policy baseline, empirically estimated opacity channel and actual licensing-approval fractions are unavailable and are not replaced by those hypothetical paths.

Reported scenario payoffs integrate 30 years without terminal continuation. They are not infinite-horizon welfare estimates and not a causal policy ranking. Physical outage/partial-access and alternative-routing tests use synthetic networks, because the empirical routes remain incomplete. ODE fold diagrams are sensitivity illustrations, not estimated industrial tipping points.

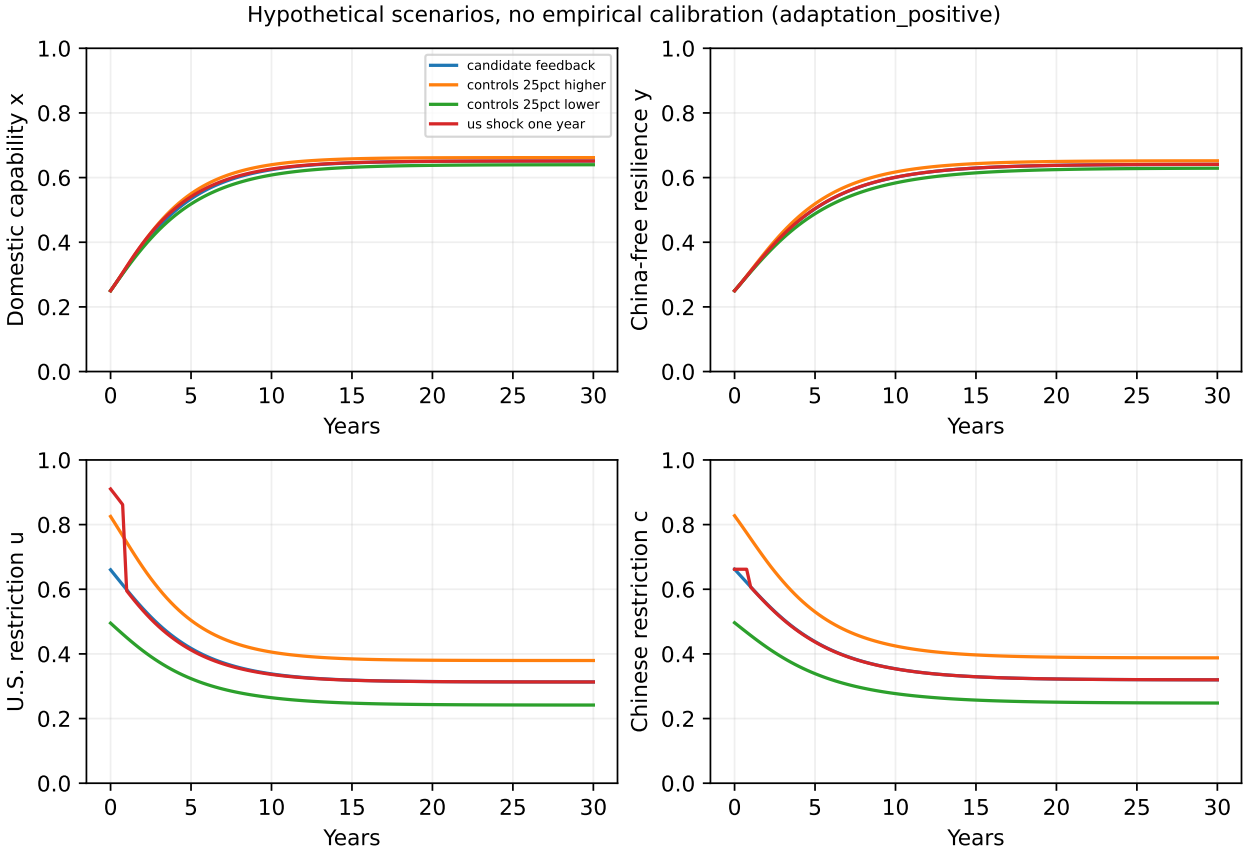


Figure 30: Hypothetical trajectories under a verified finite-grid candidate and specified control perturbations. The initial state and all parameters are scenario inputs. No trajectory is an observed U.S./China policy path or an empirically calibrated forecast.

11 Limitations

The decisive limitation is measurement coverage, not a missing optimizer. Native rare-earth output remains insufficient to identify element-specific recovered flows, input ownership/origin, transport contracts, inventories, other magnet constituents, target-grade demand and commercial qualification. The documented project subset cannot estimate population success rates; cancellations and inaccessible issuers require systematic discovery. Source reporting delay, left truncation, interval uncertainty and common shocks need richer historical records before hierarchical hazard fitting.

Semiconductor technical targets are provisional. Public investment inputs do not identify qualified manufacturing capability, and selected legal instruments do not reveal actual licenses, servicing, diversion, usable installed capital or foreign alternatives. Without credible comparison groups, anticipation/spillover checks and product-specific exposure, causal effects remain unidentified. The patent and innovation literature cannot substitute for direct engineering evidence.

Opacity obligations, expected release denominators and independent coding are missing. Treating inaccessible web pages as intentional opacity would invalidate the proposed index. Numerical game results are conditional on unestimated primitives and discretization. Pure stationary continuum existence/uniqueness is not proved, some fine-grid scenarios fail, and off-grid residuals remain nonzero. A successful synthetic recovery test is not empirical calibration.

The bibliography audit verifies varying levels of evidence; publisher-deposited abstracts do not equal full-paper replication. Some original legal documents and reports remain inaccessible. The manuscript requires independent mineral-processing, semiconductor-control and dynamic-game specialist review and independent-machine reproduction before submission. The draft has not been submitted to a journal and is not represented as publication-ready.

12 Conclusion

The executed work establishes a functioning computational research package and selected source-traceable descriptive industrial and legal evidence. It verifies a limited self-eroding-leverage theorem, conserving process optimization, historical-information safeguards and conditional numerical game candidates. It also demonstrates why announced capacity, R&D spending, licensing requirements and missing disclosure data cannot be converted directly into resilience, capability or intentional opacity.

The central empirical hypothesis remains open. The present records are consistent with investment and production changes but do not identify how restrictions caused those changes. A completed study must supply the missing industrial network, historical risk set, engineering targets and observability obligations before issuing real-world probability or policy claims. The archived failures and explicit identification gates make that unfinished boundary reproducible.

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A Technical appendices

A.1 Forward invariance and fixed-policy stability

At $m = 0$, $\dot{m} = \kappa_C u \geq 0$; at $m = 1$, $\dot{m} = \kappa_C(u - 1) \leq 0$. The same holds for n. At $x = 0$, growth is $g_C e^{\beta_C m} > 0$ for either sign of β_C; at $x = 1$, drift is $-\delta_C \leq 0$. The analogous statements hold for y. Local Lipschitz continuity on the compact state box and these inward boundary conditions give forward invariance for admissible measurable controls. This proof does not rely on post-integration clipping. With pipelines ordered first, the constant-policy Jacobian is block triangular, yielding the four negative eigenvalues reported above. Under feedback, the additional terms are $f_u \nabla\sigma_U + f_c \nabla\sigma_C$ and must be retained.

A.2 Nine continuous-action regimes

The regimes $(0,0), (1,0), (0,1), (1,1)$ require respectively

$$\begin{aligned} M_U \leq 0,\ M_C \leq 0; \quad & M_U \geq k_U,\ M_C + q_C \leq 0; \\ M_U + q_U \leq 0,\ M_C \geq k_C; \quad & M_U + q_U \geq k_U,\ M_C + q_C \geq k_C. \end{aligned}$$

The four mixed boundary/interior candidates are $(M_U/k_U, 0)$, $((M_U + q_U)/k_U, 1)$, $(0, M_C/k_C)$ and $(1, (M_C + q_C)/k_C)$, each subject to its own strict interior bound and the other player's boundary derivative inequality. For $\Delta = k_U k_C - q_U q_C \neq 0$, the interior candidate is

$$(u_I, c_I) = \left(\frac{k_C M_U + q_U M_C}{\Delta}, \frac{k_U M_C + q_C M_U}{\Delta} \right), \tag{21}$$

and is eligible only when both coordinates are strictly interior. Shared boundary solutions are deduplicated. When $\Delta = 0$, compatibility and box constraints may define a continuum rather than an isolated interior pair. Conditional Hamiltonian concavity makes best-response residuals sufficient for the statewise action game; it does not solve the coupled HJB fixed point.

A.3 Constant-commitment curvature and uniqueness

For either own-action subsystem, write $M(t) = v + (M_0 - v)e^{-\kappa t}$, $\ell(t) = 1 - e^{-\kappa t}$ and $G(t) = ge^{\beta M(t)}$. The state satisfies $\dot{z} = G(1 - z) - \delta z$. Differentiability in the constant action follows from the smooth bounded vector field. With fixed initial state, sensitivities $\xi = \partial_v z$, $\zeta = \partial_{vv} z$ start at zero and obey

$$\dot{\xi} = G\beta\ell(1 - z) - (G + \delta)\xi, \tag{22}$$

$$\dot{\zeta} = G\beta^2\ell^2(1 - z) - 2G\beta\ell\xi - (G + \delta)\zeta. \tag{23}$$

Let $G_- = ge^{\min(\beta,0)}$, $G_+ = ge^{\max(\beta,0)}$, and $\lambda_0 = G_- + \delta > 0$. Variation of constants and $0 \leq z, \ell \leq 1$ give $|\xi(t)| \leq G_+|\beta| \int_0^t e^{-\lambda_0(t-r)} dr$. Integrating against $e^{-\rho t}$ and applying Fubini to the nonnegative majorant yields

$$\int_0^\infty e^{-\rho t}|\xi(t)|dt \leq S = \frac{G_+|\beta|}{\rho(\rho + \lambda_0)}, \tag{24}$$

$$\int_0^\infty e^{-\rho t}|\zeta(t)|dt \leq R = \frac{G_+\beta^2}{\rho(\rho + \lambda_0)} + \frac{2G_+|\beta|S}{\rho + \lambda_0}. \tag{25}$$

These dominating bounds justify differentiation of W. In the player-indexed bounds, β_i denotes the adaptation coefficient of that player's own-action subsystem: β_C for the U.S. action and β_U for the Chinese action. If $\beta \geq 0$, the first sensitivity equation gives $\xi \geq 0$ and thus $W' \leq 0$. For either sign,

$$F_i'' = A_i(2W_i' + vW_i'') - k_i/\rho_i \leq A_i\{R_i + 2S_i\mathbf{1}_{\{\beta_i<0\}}\} - k_i/\rho_i = -\mu_i. \tag{26}$$

When $\mu_i > 0$, the unique constrained maximizer is characterized by the interior derivative root or the appropriate endpoint inequality. Strong monotonicity of the derivative, including constrained endpoints, bounds the best-response Lipschitz constant by $L_i = |q_i|/(\rho_i\mu_i)$. If $L_U L_C < 1$, the composition of the two best responses is a contraction on $[0, 1]$. Its unique fixed point and the associated other response give existence and uniqueness within the restricted commitment class. These are sufficient, not necessary, conditions. Their failure would not establish nonexistence.

At $\beta = 0$, let $z^* = g/(g + \delta)$. Then

$$W = \frac{1 - z^*}{\rho} + \frac{z^* - z_0}{\rho + g + \delta}. \tag{27}$$

The restricted game reduces exactly to the original simultaneous box-constrained action game with $M_i = \rho_i A_i W_i$, the same k_i, q_i. Tests compare both the exact solution and a corner case where independently clipping the interior solution fails.

For truncation at T, the omitted W tail is bounded by $e^{-\rho T}/\rho$. Uniform sensitivity bounds are $B_1 = G_+|\beta|/\lambda_0$ and $B_2 = (G_+\beta^2 + 2G_+|\beta|B_1)/\lambda_0$, giving first/second derivative tails at most $B_1 e^{-\rho T}/\rho$ and $B_2 e^{-\rho T}/\rho$. On the invariant box, an absolute full-flow-payoff bound is $A_i + D_i + k_i/2 + h_i/2 + B_i + |q_i|$, so the omitted full-return tail is bounded by this constant times $e^{-\rho_i T}/\rho_i$. The integration endpoint is chosen as $32/\min_i \rho_i$. Exact sufficient conditions, analytic tails and standard numerical quadrature checks are reported separately; no interval-arithmetic certificate or full dynamic equilibrium implication is claimed.

A.4 Monotone generator, residuals and linear recovery

For state dimension j with spacing h_j, positive drift creates a transition to the forward neighbor at rate f_j^+/h_j, and negative drift to the backward neighbor at rate f_j^-/h_j. Missing outward rates are zero under invariance. The generator L has nonnegative off-diagonals and row sum zero. Thus $\rho I - L$ is a strictly discounted M-matrix, and policy evaluation solves $(\rho_i I - L)V_i = \pi_i$. No arbitrary zero-value boundary condition is imposed.

The implemented residual is $R_i = \rho_i V_i - \pi_i - LV_i$, with tolerance $10^{-8}(1 + \|\pi_i\|_\infty)$. Own-action advantages maximize the full piecewise discrete Hamiltonian; action error tolerance is 10^{-7}. BiCGSTAB stagnation in a held-out fine-grid evaluation was resolved with a recorded GMRES fallback without relaxing tolerances. An independent direct-solve test checks the fallback. The sign-changing control drift requires a five-status-per-player enumeration: lower boundary, lower-piece interior, pipeline kink, upper-piece interior and upper boundary. Global unilateral checks reject locally stationary but suboptimal candidates.

A.5 Discounted finite-state regret comparison

Fix one opponent's feedback policy and suppress the player index. For a candidate own policy π, generator L_π, reward r_π, numerical value V and discount $\rho > 0$, define

$$E = \|\rho V - r_\pi - L_\pi V\|_\infty, \quad G = \max_s \max_a \{r_a(s) + (L_a V)(s) - r_\pi(s) - (L_\pi V)(s)\}. \tag{28}$$

The discounted Markov resolvent is positive and has sup norm $1/\rho$, so the exact policy value obeys $\|V - V_\pi\|_\infty \le E/\rho$. Every generator annihilates constants. Shifting V upward by $(G+E)/\rho$ therefore makes an optimal Bellman supersolution; comparison implies

$$0 \le V^* - V_\pi \le (G + 2E)/\rho. \tag{29}$$

For an explicit unilateral feedback policy μ, let its numerical value be W, policy residual E_μ and global Hamiltonian gap G_μ. Statewise lower and upper comparisons are

$$\ell(s) = \max\{0, W(s) - V(s) - (E + E_\mu)/\rho\}, \tag{30}$$

$$b(s) = \min\{(G + 2E)/\rho,\ \max\{0, W(s) - V(s) + (E + E_\mu + G_\mu)/\rho\}\}, \tag{31}$$

$$\ell(s) \le V^*(s) - V_\pi(s) \le b(s). \tag{32}$$

The lower comparison uses feasibility of μ and the two policy-evaluation error bounds; the upper uses the Bellman supersolution comparison for W. Reported global endpoints are $\max_s \ell(s)$ and $\max_s b(s)$. The argument concerns a discounted finite-state continuous-action control problem with the opponent held fixed; the original generator is a Markov-chain approximation, not the literal deterministic ODE. Compact actions and finite bounded rates permit the discounted Bellman comparison. The inequalities are exact-arithmetic results. Evaluating their inputs with standard sparse floating-point arithmetic does not give directed-rounding or interval-certified numerical endpoints, and these are not empirical confidence intervals.

A.6 Persistence, censoring and uncertainty

Define $\tau_h = \inf\{q : Y_{q+i} \ge h,\ i = 0, 1, 2, 3\}$. Recognition is $\tau_h + 3$. A simulation reporting start-by-T must extend through $T + 3$; otherwise the event is right-censored. Independent Monte Carlo draws have conditional error $\sqrt{\hat{p}(1-\hat{p})/N}$; 10,000 draws give a maximum of 0.005 under independence. This error is not parameter, structural or measurement uncertainty. No real-world credible interval follows from the synthetic run.

The first-exit likelihood uses the probability of remaining at risk for observed complete intervals, followed by a cause-specific exit or right censor. Interval observations sum admissible exit probabilities rather than assigning an arbitrary event date. This restricted observation model is not a substitute for a hidden multistate model that permits unobserved repeated transitions. Quarterly expansion cannot manufacture information about unobserved projects.

A.7 Physical integrity and unit conventions

Activities convert explicitly represented product forms; oxide stoichiometry uses molecular weights and separate recovery coefficients. Contained Nd, Pr, Dy and Tb do not appear from processing. Magnet recipes preserve coupled element requirements; finite Fe/B inputs in richer fixtures prevent hidden unconstrained supply. Recovery loss and disposal are accounted for explicitly. All-stage supplier failures remove linked origin-tagged feeds, owned facilities, transport rights and inventories as defined by the registered scenario. Multi-quarter balances prevent repeated use of a one-time stockpile. The stateless simulator bridge rejects networks with intertemporal inventory rather than silently reusing it.

A.8 Replication and acceptance boundary

The reproduction entry point is `scripts/replicate.py`; `--full` recomputes the large game solves as well as empirical extraction, validation, scenarios and figures. The default verifies saved large-game arrays independently. The environment has a resolved dependency lock and a digest-pinned container base. Source checksums, configurations, seeds, command exit statuses, figure hashes and manuscript claim links are recorded in generated manifests. A clean container on the same host is distinct from independent-machine replication.

The acceptance ledger contains each Section 14 matrix requirement plus detailed implementation and release conditions, with VERIFIED, PARTIAL, BLOCKED or NOT ATTEMPTED status and file evidence. A passing mathematical fixture cannot upgrade an empirical-data requirement. Precise remaining requirements are listed in `audit/remaining_requirements.md`. The manuscript and public-facing research status must remain evidence-limited until those requirements are met.